[450] Note on the Theory of the Rational Transformation between Two Planes, and on Special Systems of Points. *(concluded)*

(Continuation from the previous chunk; closing pages of paper 450.*)*

I believe the better course is to assume (1) and (3) as the fundamental equations, from them deducing (2); and we thus also get over a difficulty presently referred to, but which did not occur to me when the memoir was written.

In fact, starting with the equations

$$x' : y' : z' = \frac{X}{x} : \frac{Y}{y} : \frac{Z}{z}$$

(which are to give $x : y : z = X' : Y' : Z'$), we have in the first instance the equation (1). Moreover, establishing for x', y', z' a linear equation $ax' + by' + cz' = 0$, we have corresponding hereto a curve $aX + bY + cZ = 0$, and the coordinates x, y, z of a point on this curve are proportional to $X' : Y' : Z'$; that is, substituting for z' the value $-\dfrac{1}{c}(ax' + by')$, they are proportional to rational and integral (homogeneous) functions of (x', y'), that is, to rational and integral functions of the single parameter $x' : y'$; wherefore the curve $aX + bY + cZ = 0$ is unicursal; whence the equation (3). The like change may be made in the theory of the rational transformation between two spaces; and it is in this case a more important one.

The difficulty is as follows: It is not self-evident that we are at liberty to assume

$$a_1 + 3a_2 + 8a_3 + \cdots \leq \tfrac{1}{2}(n^2 + 3n) - 2;$$

for imagine that we had a system of $(a_1, a_2, a_3, \ldots)$ points, such that $a_1 + 4a_2 + \cdots$ being $= n^2 - 1$, and $a_1 + 3a_2 + \cdots$ being $> \frac{1}{2}(n^2 + 3n) - 2$, the points were such that the conditions in question (viz., the condition that the curve passes once through each of the points a_1, twice through each of the points a_2, ...) should be less than $a_1 + 3a_2 + \cdots$, and in fact $=$ or $< \frac{1}{2}(n^2 + 3n) - 2$; then the functions X, Y, Z would not of necessity be connected by a linear relation $\lambda X + \mu Y + \nu Z = 0$, and the ground for the assumption in question, $a_1 + 3a_2 + \cdots < \frac{1}{2}(n^2 + 3n) - 2$, would no longer exist. And except by the process now adopted of deriving the equation (2) from the equations (1) and (3), I do not know how the impossibility of such a system is to be established; viz., I do not know how we are to prove the following theorem: *There is not any system of* $(a_1, a_2, a_3, \ldots)$ *points*, where

$$a_1 + 4a_2 + 9a_3 + \cdots = n^2 - 1,$$
$$a_1 + 3a_2 + 6a_3 + \cdots > \tfrac{1}{2}(n^2 + 3n) - 2,$$

such that (for a curve of the order n passing once through each point a_1, twice through each point a_2, ...) the number of conditions actually imposed on the curve is $=$ or $< \frac{1}{2}(n^2 + 3n) - 2$.

A system of $(a_1, a_2, \ldots)$ points such that the number of actually imposed conditions is less than $a_1 + 3a_2 + \cdots$, may be termed a *special* system; we have, of course, the well-known case $(a_1 = n^2)$ of a system of n^2 points, such that any curve of the order n passing through $\frac{1}{2}(n^2 + 3n) - 1$ of these passes through all the remaining points {or what is the same thing, where the number of conditions actually imposed is $= \frac{1}{2}(n^2 + 3n) - 1$}; and we have the following special system, which presented itself to Dr Clebsch, in his researches on the *Abbildung* of a quintic surface with two non-intersecting nodal lines; viz., "we may have $a_1 = 12$, $a_2 = 2$. We may have 12 points and 2 points such that, for a quintic curve passing once through each of the 12 points and twice through each of the 2 points, the number of conditions actually imposed (instead of being $12 + 3 \cdot 2, = 18$) is $= 17$." The construction is as follows: viz., starting with the 2 points and any 10 points, we may draw a quartic passing twice through the first of the 2 points, once through the second of them, and through the 10 points; and another quartic passing twice through the second of the 2 points, once through the first of them, and through the 10 points: the two quartics intersect in the 2 points each twice, in the 10 points, and in 2 new points, forming, with the 10 points, a system of 12 points; and the first-mentioned 2 points and the 12 points form the system in question.

A more complicated case, $a_1 = 10$, $a_2 = 6$, $a_3 = 1$, occurs in Dr Nöther's paper, "Ueber Flächen, welche Schaaren rationaler Curven besitzen," [*Math. Ann.*, t. iii. (1871), pp. 161–227]. Except these two, I do not know any other case of a special system for which $a_2, a_3, \ldots$ are not all $= 0$; the investigation of such systems would, I think, be very interesting.

[A concluding paragraph of seven lines gave some corrections to the "Memoir on the Rational Transformation between Two Spaces," 447, which corrections are made in the present reprint of that paper.]

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[451] A Second Memoir on Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 198–202. *Read December* 8, 1870.*]*

In my Memoir on Quartic Surfaces, *ante* pp. 19–69, [445], although remarking (see No. 79) that the identification was not completely made out, I tacitly assumed that the symmetroid and the decadianome (each of them a quartic surface with 10 nodes) were in fact identical. There is yet a good deal which I cannot completely explain; but the truth appears to be, that the decadianome includes two cases of coordinate generality, say the *sextic decadianome*, and the *bicubic decadianome* = *symmetroid*: viz., in the first of these the circumscribed cone, having for vertex any one of the 10 nodes, is a proper sextic cone with 9 double lines; in the second it is a system of two cubic cones, intersecting, of course, in 9 lines, which are double lines of the aggregate sextic cone: or, in the notation of the Table No. 11, in the case of the sextic decadianome, the circumscribed cones are each of them 6_9; in that of the bicubic decadianome = symmetroid, they are each of them $(3, 3)$. We thus arrive at a very remarkable system of 10 points in space, viz., giving the name "ennead" to any 9 points *in plano*, which are the intersections of two cubic curves, or to any 9 lines through a point which are the intersections of two cubic cones; the 10 points in space are such that, taking any one whatever of them as vertex, and joining it with the remaining points, the 9 lines form an ennead. I purpose in the present short Memoir to consider the theories in question; the paragraphs are numbered consecutively with those of the Memoir on Quartic Surfaces.

Plane Sextic Curve with 9 Nodes.

110. A sextic curve contains 27 constants; and the number of conditions to be satisfied in order that a given point may be a node is = 3. Hence it would at first sight appear that the curve could be found so as to have 9 given nodes; this would be $9 \times 3, = 27$ conditions, or the curve would be completely determinate. But observe that through the 9 given points we have a determinate cubic curve $U = 0$; we have therefore U^2 = a sextic curve, and the only sextic curve with the 9 given nodes; that is, there is not in a proper sense any sextic curve with the 9 given nodes. The number of given nodes is thus = 8 at most.

111. The sextic curve with 8 given nodes should contain $27 - 3 \cdot 8 = 3$ constants. We may through the 8 given points draw the two cubics $P = 0$, $Q = 0$; and we have then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve having the 8 nodes, and also a ninth node, viz., the remaining point of intersection of the two cubic curves, or say the remaining point of the ennead. Hence if $V = 0$ be any particular sextic curve having the 8 given nodes, we have

$$(a, b, c \mid P, Q)^2 + \theta V = 0,$$

a proper sextic curve having the 8 given nodes; and this, as containing the right number ($= 3$) of constants, will be the general sextic curve having the 8 given nodes.

112. There will be a ninth node if $\theta = 0$; viz., the curve is then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve, having for nodes the 9 points of the ennead. Observe that the ninth node is here a point completely and uniquely determined by means of the given 8 nodes. Moreover the number of constants is $= 2$, so that we have here a general (improper) solution of the question of finding a sextic curve with 9 nodes, 8 of them given.

113. But if θ is not $= 0$, then the ninth node must be a point on the curve $J(P, Q, V) = 0$; viz., this is a curve of the order 9, determined by means of the given 8 points; say it is the "dianodal curve" of these 8 points, and, as is easy to see, it has each of these 8 points for a node. The ninth node of the sextic may be any point whatever on the dianodal curve; and regarding it as a given point, the sextic will still contain 1 constant; that is, we have the general solution of the problem of finding a sextic curve with 9 nodes, 8 of them given, and the 9th a given point on the dianodal curve.

114. So long as the 8 points are arbitrary, the dianodal curve does not pass through the 9th point of the ennead, and the two cases above considered are mutually exclusive. It will be noticed how closely analogous this theory of the plane sextic with 9 nodes, is to that of the quartic surface with 8 nodes.

115. Of course, instead of the plane sextic curve, we may have a sextic cone; such a cone has at most 8 given double lines; and if there be a 9th double line, then there are the two cases of coordinate generality; viz., (1), the new double line is the ninth line of the ennead, the cone being in this case not a proper sextic cone, but a bicubic cone; (2), the new double line may be any line whatever on the dianodal cone, (cone of the order 9 determined by the 8 given lines, and having each of these for a double line,) and regarding it as a given line on the dianodal cone, the sextic cone contains 1 constant.

Each circumscribed cone of the Symmetroid is $(3, 3)$.

116. Using (x, y, z, w) as current coordinates of a point of the symmetroid, I take S, T, U, V to be quadric functions of the coordinates $(\alpha, \beta, \gamma, \delta)$; the equation of the symmetroid is therefore given by

$$xS + yT + zU + wV = \text{cone},$$

and the nodes thereof are determined by

$$xS + yT + zU + wV = \text{plane-pair}.$$

Suppose that a node is $(x = 0, y = 0, z = 0)$; the condition for this is $V =$ plane-pair, and we may without loss of generality write $V = \gamma^2 + \delta^2$. Hence, putting for shortness $\Omega = xS + yT + zU$, that is a quadric function

$$(a, b, c, d, f, g, h, l, m, n \,|\, \alpha, \beta, \gamma, \delta)^2,$$

wherein the coefficients $a, b, \ldots$ are arbitrary linear functions of (x, y, z), but not containing w, the equation of the symmetroid is given by

$$\Omega + w(\gamma^2 + \delta^2) = \text{cone}.$$

117. It follows that the equation is

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c+w, & n \\ l, & m, & n, & d+w \end{vmatrix} = 0;$$

viz., this is

$$V + w(\partial_c + \partial_d)V + w^2\partial_c\partial_d V = 0,$$

where V denotes the foregoing determinant, writing therein $w = 0$. Or, observing that V contains c and d each only linearly, the equation may be written

$$V + w(\partial_c V + \partial_d V) + w^2\partial_c\partial_d V = 0,$$

which is a quartic surface having, as it should have, the point $(0, 0, 0)$ for one of its ten nodes.

118. The equation of the circumscribed cone is

$$(\partial_c V - \partial_d V)^2 + 4(\partial_c V \cdot \partial_d V - V \cdot \partial_c\partial_d V) = 0,$$

or, what is the same thing, it is

$$(\partial_c V - \partial_d V)^2 + 4(\partial_c V \cdot \partial_d V - V \cdot \partial_c\partial_d V) = 0.$$

But we have identically

$$\partial_c V \cdot \partial_d V - (\tfrac{1}{2}\partial_{cd} V)^2 = V \cdot \partial_c\partial_d V;$$

so that the equation is

$$(\partial_c V - \partial_d V)^2 + (\partial_{cd} V)^2 = 0;$$

a sextic cone breaking up into the two cubic cones

$$\partial_c V - \partial_d V \pm i\,\partial_{cd} V = 0,$$

so that the cone is $(3, 3)$. And since clearly the point $(0, 0, 0)$ may be regarded as representing any one whatever of the 10 nodes, it follows that for any node whatever of the symmetroid, the circumscribed cone is $(3, 3)$, so that, as stated above, bicubic decadianome = symmetroid.

Deductions from the foregoing theory.

119. Referring to No. 85 of the original memoir, it appears that, with 6 given points as nodes, we can actually find for the symmetroid an equation containing 6 constants. I cannot discover any ground for doubting that 3 of these may be determined so as to give to the symmetroid a seventh given node; and I therefore assume that with 7 given points as nodes, an equation can be found with 3 constants. The symmetroid is certainly not octadic, hence the eighth node must lie on the dianodal surface of the 7 given points. I can discover no ground for doubting but that two of the constants may be determined so that the eighth node shall be any given point whatever on the dianodal surface of the 7 points; and (this being so) that further the remaining constant may be determined so that the ninth node shall be any given point whatever on the dianodal curve of the 8 points. But if all this be so, the consequence is very remarkable; the tenth node is not any one whatever of the 22 dianodal centres of the 9 points, but it is a uniquely determinate "enneadic centre," viz., we must have the following theorem:

120. "Take any 7 points; an eighth point at pleasure on the dianodal surface of the 7 points; a ninth point at pleasure on the dianodal curve of the 8 points. In the system of 9 points so determined, take any one as vertex, and joining it with the remaining 8, construct the ninth line of the ennead. Performing this construction with each of the 9 points successively as vertex, we obtain 9 lines passing through the 9 points respectively. These 9 lines meet in a point which is the 'enneadic centre' of the 9 points: and further, the 10 points form a completely symmetrical system, so that each one of them is the enneadic centre of the remaining 9."

121. Assuming that the 9 lines do intersect so as to give rise to an enneadic centre, there is no difficulty in conceiving that the loci, which by their intersection determine the dianodal centres, do each of them pass through the enneadic centre; so that this enneadic centre counts once or more among the dianodal centres, and the number of proper dianodal centres, instead of being $= 22$, will be suppose $= 22 - \omega$; and if, further, the 9 points, together with the enneadic centre, are the nodes of a symmetroid, but the 9 points together with any one of the $22 - \omega$ dianodal centres are the nodes of a sextic decadianome, then we must also have as follows:

122. "Considering any 9 points as above; taking any one as vertex, and joining it with the remaining 8, these 8 lines determine a dianodal ninthic cone. We have thus 9 dianodal cones, which cones pass all of them through the same $22 - \omega$ points."

123. I am not able to verify these theorems *à posteriori*. It appears to me that the theorem in regard to the enneadic centre subsists for a system of 9 points such as referred to in the statement; but that if by possibility the statement be too general, the theorem must, at all events, subsist for a more special system of 9 points; and that there certainly exist systems of 10 points, such that each 9 of the points have as an enneadic centre the tenth point. (*I have since ascertained that if a quartic surface with* 10 *nodes has a single node* $(3,3)$, *the surface is a symmetroid; whence, by what precedes, the remaining nine nodes are each of them* $(3,3)$. *Added* 25 *March,* 1871.)

124. I notice, as a subject of investigation, the following system of correspondence; viz., given any 8 points in space: then to every point in space corresponds a line through this point, viz., the ninth line of the ennead obtained by joining the point with the 8 given points respectively; and to each line in space a point or points on the line, viz., the point or points for each of which the line is the ninth line of the ennead obtained by joining the point with the eight given points respectively.

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[452] On an Analytical Theorem from a New Point of View.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 220, 221. *Read February* 9, 1871.*]*

The theorem is a well-known one, derived from the equation

$$(az^2 + 2bz + c)w^2 + 2(a'z^2 + 2b'z + c')w + a''z^2 + 2b''z + c'' = 0;$$

viz., considering this equation as establishing a relation between the variables z and w, and writing it in the forms

$$0 = u = Aw^2 + 2Bw + C = A'z^2 + 2B'z + C' = 0,$$

(where, of course, A, B, C are quadric functions of z, and A', B', C' quadric functions of w,) we have

$$0 = \frac{du}{dw}\,dw + \frac{du}{dz}\,dz = (Aw + B)\,dw + (A'z + B')\,dz;$$

but in virtue of the equation $u = 0$, we have $Aw+B = \sqrt{B^2 - AC}$, and $A'z+B' = \sqrt{B'^2 - A'C'}$, and the differential equation thus becomes

$$\frac{dw}{\sqrt{B'^2 - A'C'}} + \frac{dz}{\sqrt{B^2 - AC}} = 0,$$

where $B'^2 - A'C'$ and $B^2 - AC$ are quartic functions of w and z respectively. This is, of course, integrable (viz., the integral is the original equation $u = 0$); and it follows, from the theory of elliptic functions, that the two quartic functions must be linearly transformable into each other; viz., they must have the same absolute invariant $I^3 \div J^2$. It is, in fact, easy to verify, not only that this is so, but that the two functions have the same quadrinvariant I, and the same cubinvariant J.

The new point of view is, that we take the coefficients a, b, &c., to be homogeneous functions of (x, y), their degrees being such that the equation $u = 0$ is a quartic equation $(*\,|\,x, y, z, w)^4 = 0$: viz., this equation now represents a quartic surface having a node (conical point) at the point $(x = 0, y = 0, z = 0)$, and also a node at the point $(x = 0, y = 0, w = 0)$, say, these points are O, O' respectively. The equation $B^2 - AC = 0$ gives the circumscribed sextic cone having O for its vertex, and the equation $B'^2 - A'C' = 0$ the circumscribed sextic cone having O' for its vertex; each of these cones has the line OO' $(x = 0, y = 0)$ for a nodal line, as appears geometrically, and also by the equations containing z, w respectively in the degree 4. Considering $B^2 - AC$ as a quartic function of z, its quadrinvariant is a function $(x, y)^4$, and its cubinvariant a function $(x, y)^6$; and similarly, considering $B'^2-A'C'$ as a quartic function of w, its invariants are functions $(x, y)^4$ and $(x, y)^6$. We have thus, between the two cones, a geometrical relation answering to the analytical one of the identity of the invariants; but the nature of this geometrical relation is not obvious; and it presents itself as an interesting subject of investigation.

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[453] On a Problem in the Calculus of Variations.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 221, 222. *Read February* 9, 1871.*]*

The problem is, $z = \frac{1}{2}(3x - y^2)\,y$, to find y a function of x such that $\int z\,dx =$ max. or min., subject to a given condition $\int y\,dx = c$ (the limits of each integral being x_0, x_1, where these quantities are each positive, and $x_1 > x_0$). The ordinary method of solution gives $y^2 = x + \lambda$, where $(x_1 + \lambda)^{3/2} - (x_0 + \lambda)^{3/2} = \frac{3}{2}c$; so long as c is not less than $(x_1 - x_0)^{3/2}$, there is a real value of λ, but for a smaller value of c there is no real value. The difficulty arising in this last case is somewhat illustrated by replacing the original problem by a like problem of ordinary maxima and minima; viz., $x_1, x_2, \ldots x_n$ being given positive values of x, in the order of increasing magnitude; and if, in general, $z_i = \frac{1}{2}(3x_i - y_i^2)\,y_i$, then the problem is to find y_i a function of x_i, such that $\Sigma z_i =$ max. or min., subject to the condition $\Sigma y_i = c$. We have here $y_i^2 = x_i + \lambda$ where λ is to be determined by the condition $\Sigma y_i = c$; the remainder of the investigation turns on the question of the sign $y_i = +\sqrt{x_i + \lambda}$ or $y_i = -\sqrt{x_i + \lambda}$, to be taken for the several values of i respectively.

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[454] A Third Memoir on Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 234–266. *Read April* 13, 1871.*]*

The present Memoir is a continuation of my former researches on Nodal Quartic Surfaces, [445, 451]. The leading idea is, that for a quartic surface with k-nodes, given the nature of the circumscribed, $(k-1)$ nodal, sextic cone belonging to any one node of the surface (for instance, $k=10$, that it is a cone $(3,3)$ composed of two cubic cones), we thereby determine the equation of the quartic surface, and consequently the nature of the remaining $(k-1)$ nodes thereof. By means of this general theory I complete, in an essential point, the theory of the Symmetroid; viz., I show that a 10-nodal quartic surface having a single node $(3,3)$ is a Symmetroid; whence, as appears by my second Memoir, [451], each of the remaining nine nodes is also a node $(3,3)$; and we have the theory of the remarkable system of ten points in space such that, joining any one of them with the remaining nine, the nine lines thus obtained are the intersections of two cubic cones. A large part of the Memoir is devoted to the consideration of the surfaces with 16, 15, 14, and 13 nodes: this is substantially a reproduction of the results obtained by Kummer in the Memoir "Ueber die algebraischen Strahlensysteme, &c.," already referred to; but the results in question are brought into connexion with the theory of the present Memoir, and they are, by a change of the constants, exhibited in a form of much greater symmetry and elegance. I attach importance also to the square diagrams by means of which I have exhibited, in a compendious form, the relation between the several nodes and circumscribed sextic cones.

The paragraphs are numbered consecutively with those of the first and second Memoirs.

Preliminary Considerations and Classification.

125. I call to mind that if a quartic surface has a node (conical point), then there is for this node a tangent quadricone and a circumscribed sextic cone; viz., if the surface has $(k-1)$ other nodes, or in all k nodes, then the sextic cone has $(k-1)$ nodal lines (passing through the other nodes respectively), and we have thus for the different forms of the sextic the table No. 11; viz., this is

Circumscribed Sextic Cone.

Nodes of Surface	Nodal Lines of Cone	Forms
1	0	6
2	1	6_1
3	2	6_2
4	3	6_3
5	4	6_4
6	5	$6_5,\ 5$
7	6	$6_6,\ 5_1$
8	7	$6_7,\ 5_2,\ 4,\ 2$
9	8	$6_8,\ 5_3,\ 4_1,2,\ 4,1,1$
10	9	$6_9,\ 5_4,\ 4_2,2,\ 4_1,1,1,\ 3,3$
11	10	$6_{10},\ 5_5,\ \ldots,\ 4_3,1,1,\ 3_1,3$
12	11	$5_6,\ \ldots,\ 4_3,1,1,\ 3,2,1$
13	12	$\ldots,\ 3,2,1,\ 3,1,1,1,\ 2,2,2$
14	13	$\ldots,\ 3_1,1,1,1,\ 2,2,1,1$
15	14	$\ldots,\ 2,1,1,1,1$
16	15	$1,1,1,1,1,1$

viz., 6 denotes a proper sextic cone without nodal lines; 6_1 a proper sextic cone with one nodal line; $5, 1$ a proper quintic cone and a plane, &c.

We may distinguish the nodes according to the sextic cones; thus, a node 6 means a node for which the circumscribed cone is a proper sextic cone, $(1, 1, 1, 1, 1, 1)$ a node where the circumscribed cone breaks up into six planes, &c.

126. A 16-nodal surface has 16 nodes $(1, 1, 1, 1, 1, 1)$, and a 15-nodal surface has 15 nodes $(2, 1, 1, 1, 1)$; but, for a 14-nodal surface, the question arises how many nodes are $(3_1, 1, 1, 1)$, and how many $(2, 2, 1, 1)$. It was remarked, No. 13, that the only possible cases were $14, 0$; $8, 6$; or $2, 12$; and that we might, in like manner, limit the number of possible cases for other values of k; but that the inquiry was not then further pursued. I resume this inquiry, but without obtaining as yet a complete answer.

127. It is to be observed that a line joining any two nodes is not, in general, a line on the surface, but that it may be so; the surfaces for which this is so (viz., any surface which contains upon it a line through two nodes) form, however, a class by themselves, which at present I altogether exclude from consideration. This being so, it will appear in the sequel that there is but one kind of surface having a node $(2, 2, 1, 1)$, and but one kind of surface having a node $(3_1, 1, 1, 1)$. Now there is a surface, Kummer's 14-nodal, the nodes of which are $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$; wherefore the two kinds are identical, and are each of them Kummer's 14-nodal surface. Similarly, for the 13-nodal surfaces, there is but one kind having a node $(4_1, 1, 1)$, but one kind having a node $(3, 1, 1, 1)$, and moreover but one kind having a node $(3_1, 2, 1)$; and we have Kummer's 13-nodal surface with the nodes $3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$; hence the three kinds are identical with each other and with Kummer's. Moreover, there is but one kind having a node $(2, 2, 2)$; hence all the other nodes must be $(2, 2, 2)$, and we have a surface $13(2, 2, 2)$ not given by Kummer. And in like manner for the 12-nodal surfaces, we have the two kinds given by Kummer, and a third kind $12(4_1, 1, 1)$ not given by him; the arrangement thus far being

No. of Nodes.	**Character of Surface.**
16	$16(1, 1, 1, 1, 1, 1)$,
15	$15(2, 1, 1, 1, 1)$,
14	$8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$,
13 (α)	$3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$,
(β)	$13(2, 2, 2)$.
12 (α)	$12(4_1, 2)$,
(β)	$2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$,
(γ)	$12(4_1, 1, 1)$.

128. But in the next following case we have Kummer's surface, viz.

$$11\,(\alpha) \quad 1(6_{10}) + 10(3_1, 3),$$

and I do not know whether one, two, or three kinds of surface having nodes $(4_1, 1, 1)$, $(4_2, 2)$, and $(5_5, 1)$. And in the next case we have (as will appear) the Symmetroid, viz.,

$$10\,(\alpha) \quad 10(3, 3),$$

and I do not know how many kinds of surfaces having a node or nodes 6_9, $(5_4, 1)$, $(4_2, 2)$, $(4, 1, 1)$.

It will be observed that the present division has nothing to do with the octadic and disinodal division in the former Memoir.

129. I consider a conic $A = 0$, and any six tangents thereof, $t_1 = 0$, $t_2 = 0$, $t_3 = 0$, $t_4 = 0$, $t_5 = 0$, $t_6 = 0$; we have an identical equation which might be written $AC - B^2 = t_1t_2t_3t_4t_5t_6$, but it will be convenient, introducing a constant factor K, to write it

$$AC - B^2 = K\, t_1t_2t_3t_4t_5t_6,$$

B being a cubic function and C a quartic function of the coordinates.

Consider now the series of factors, such as

$$\begin{aligned} &lA + mt_1t_2,\\ &sA + mt_1t_2t_3,\\ &UA + mt_1t_2t_3t_4,\\ &VA + mt_1t_2t_3t_4t_5,\\ &WA + mt_1t_2t_3t_4t_5t_6, \end{aligned}$$

where m is a constant, l a constant, s a linear function, U, V, W functions of the degrees 2, 3, 4 respectively; and compose with one or more such factors an expression involving the term $Kt_1t_2t_3t_4t_5t_6$; for instance, such an expression is

$$\frac{mm'}{K}\,(sA + mt_1t_2t_3)(l'A + m't_4t_5t_6);$$

this is, of the form $A\Omega + Kt_1t_2t_3t_4t_5t_6$, viz., $A\Omega + (AC - B^2)$, or $A(\Omega + C) - B^2$, say $A\Gamma - B^2$; or what is the same thing, introducing a new coordinate w, we have a quadric function

$$Aw^2 + 2Bw + \Gamma,$$

the discriminant of which, $A\Gamma - B^2$, is equal to the expression in question.

130. In the sequel (x, y, z, w) are considered as the coordinates of a point in space; $A = 0$ is thus a quadric cone, $t_1 = 0$, $t_2 = 0$, $\ldots$, $t_6 = 0$ any six tangent planes thereof; and hence $Aw^2 + 2Bw + \Gamma = 0$ is a quartic surface, having the point $(x = 0, y = 0, z = 0)$ for a node, whereof the circumscribed cone $A\Gamma - B^2 = 0$ breaks up in the assumed manner.

Thus, in order that the circumscribed cone may be as above

$$(sA + mt_1t_2t_3)(l'A + m't_4t_5t_6),$$

we have only to assume

$$\Gamma = C + \frac{mm'}{K}(sl'A + sm't_4t_5t_6 + l'mt_1t_2t_3),$$

and so in other cases. Observe that $sA + mt_1t_2t_3 = 0$ is a cubic cone, which, so long as m, s are arbitrary, has no nodal line; but establishing a single relation (say s remains arbitrary, but a proper value is assigned to m) it will be a cubic cone having a nodal line. And so $UA + mt_1t_2t_3t_4 = 0$ is a quartic cone without any nodal line, but by particularising the constants it may be made to have one, two, or three nodal lines. Such nodal determinations are obviously required in order that the formula may extend to all the before-mentioned forms of the circumscribed cone. The foregoing analysis is the foundation of the whole theory: I have given it, as above, apart from the theory, in order that the nature of it may be the better perceived; but I have now to bring it into connexion with the theory.

On the Sextic Curves, $AC - B^2 = 0$.

131. I revert to the consideration of plane curves. The equation of a sextic curve $(*\,|\,x, y, z)^6 = 0$ cannot be in general expressed in the form $AC - B^2 = 0$, where the degrees of A, B, C are 2, 3, 4 respectively; in fact, the existence of such a form implies that there is a conic $A = 0$ touching the sextic 6 times; and since a conic can only be made to satisfy 5 conditions, there is not in general any such conic.

132. Such conic, when it exists, is said to be *inscribed in* the sextic, and the sextic to be *circumscribed about* the conic, or to be an "amphigram;" and then, $A = 0$ being the equation of the conic, that of the sextic is expressible in the form in question $AC - B^2 = 0$. It is clear that $B = 0$ is a cubic curve passing through the 6 points of contact of the conic with the sextic, and that any such curve may be taken for the curve B; in fact if a particular cubic through the 6 points is $B = 0$, and the equation of the sextic is $AC - B^2 = 0$, then taking p an arbitrary linear function of the coordinates, the equation of the general cubic is $B' = B + pA = 0$; and then writing

$$\begin{aligned} A &= A, \\ B' &= B + pA, \\ C' &= C + 2Bp + Ap^2, \end{aligned}$$

we have $AC' - B'^2 = AC - B^2$; so that the original form $AC - B^2 = 0$ becomes $AC' - B'^2 = 0$. But the cubic $B = 0$ being assumed at pleasure, the quartic $C = 0$ is a determinate curve.

133. It is to be observed that a sextic curve may be an amphigram in more than one way: certainly in two, three, or four, and possibly in a greater number of ways. For the equation of the curve contains 27 constants, and hence determining the sextic so as to touch 4 given conics each of them 6 times, there are still 3 constants; and the curve will be an amphigram in regard to each of the 4 conics; say it is a quadruple amphigram. But in the sequel we are only concerned with a sextic curve considered as an amphigram in regard to a given conic $A = 0$ (no attention being paid to the other inscribed conics, if any); and then, by what precedes, taking $B = 0$ any cubic whatever through the 6 points of contact, we have a determinate quartic curve $C = 0$, and the equation of the sextic curve assumes the form $AC - B^2 = 0$.

134. The curves $A = 0$, $B = 0$, $C = 0$ contain respectively 5, 9, 14 constants; whence considering the function B as containing an arbitrary constant factor, for the curve $AC-B^2 = 0$, the number of constants is *primâ facie* $5+9+14+1 = 29$; but on account of the arbitrary linear function p, the real number is $29 - 3 = 26$: this is right, for a sextic curve contains 27 constants; and the curve being an amphigram, there is one relation between the constants, $27 - 1 = 26$.

135. Suppose now that the sextic curve $AC - B^2 = 0$ breaks up into two or more separate curves, say into the two curves $P = 0$, $Q = 0$ of the orders f, g respectively ($f + g = 6$). We have

$$AC - B^2 = PQ = 0;$$

and the conic $A = 0$ touching the sextic six times, must, it is clear, touch the curves $P = 0$, $Q = 0$, f and g times respectively. And so when the sextic breaks up into any number of curves, each component curve $P = 0$ of the order f must touch the sextic f times.

136. It follows that if the sextic break up into six lines, say $AC - B^2 = t_1t_2t_3t_4t_5t_6 = 0$, then that each of the lines $t_1 = 0$, $t_2 = 0$, $\ldots$, $t_6 = 0$ is a tangent to the conic. And conversely, starting with the conic $A = 0$ and any six tangents thereof $t_1 = 0$, $t_2 = 0$, $\ldots$, $t_6 = 0$, we have an identity of the form in question. In fact, taking any two of the tangents, say $t_1 = 0$ and $t_2 = 0$, then, if $p = 0$ be the equation of the line joining their points of intersection, the equation of the conic will be of the form $t_1t_2 + p^2 = 0$, that is, we may write $A = t_1t_2 + p^2$, or what is the same thing, $t_1t_2 = A - p^2$. (Considering A as a given quadric function of the coordinates, this of course implies that the implicit constant factors of t_1, t_2, p are properly determined.) Similarly, $q = 0$ being the line through the points of contact of t_3, t_4, and $r = 0$ that through the points of contact of t_5, t_6, we have $t_3t_4 = A - q^2$ and $t_5t_6 = A - r^2$; whence, to satisfy the equation

$$AC - B^2 = t_1t_2t_3t_4t_5t_6,$$

we have only to assume $B = lA + pqr$, l an arbitrary linear function of the coordinates, and the equation then gives

$$C = l^2A - A(p^2 + q^2 + r^2) + (q^2r^2 + r^2p^2 + p^2q^2) + l^2A + 2l\,pqr.$$

137. It will be observed that the grouping of the six tangents into pairs is arbitrary. By altering this grouping, we merely alter the linear function l, but do not obtain any new solution. Thus, say that the new form is $B' = l'A + p'q'r'$, then, by properly determining the linear function l, we can reduce this to the original form $B = lA+pqr$; viz., we can satisfy identically the equation $(l - l')A + pqr - p'q'r' = 0$; or what is the same thing, $\lambda A + pqr - p'q'r' = 0$, where λ is a linear function of the coordinates. We have, in fact, the conic $A = 0$ and the cubic $pqr = 0$ intersecting in the six points of contact; any other cubic through these six points; and consequently the cubic $p'q'r'$ must be expressible in the form $\lambda A + pqr = 0$, and we have thus the identity in question.

138. We have just seen that the value of B is necessarily of the form $B = lA + pqr$, but we are not concerned with its expression in this particular form. What we require in the sequel is a value of B, and thence one of C, satisfying the identity in question, $AC - B^2 = t_1t_2t_3t_4t_5t_6$; or what is the same thing, introducing for convenience a constant factor K, the identity

$$AC - B^2 = Kt_1t_2t_3t_4t_5t_6.$$

139. Instead of C, I write Γ, and consider the sextic amphigram $A\Gamma - B^2 = 0$ touched by the conic $A = 0$ in the points of contact of the conic with the six tangents $t_1 = 0$, $t_2 = 0$, $\ldots$, $t_6 = 0$. Suppose the sextic curve breaks up into factors; if one of these factors is a line, it is one of the six tangents, say the tangent $t_6 = 0$. If there is a conic factor, this is a conic touching the conic $A = 0$ at its points of contact with two of the tangents, say the equation is $lA+mt_1t_2 = 0$. Similarly, if there is a cubic, quartic, or quintic factor, then the equation hereof is $sA+mt_1t_2t_3 = 0$, $UA+mt_1t_2t_3t_4 = 0$, or $VA+mt_1t_2t_3t_4t_5 = 0$. Or going on to the next case of a sextic factor (being of course the whole curve), we may say that this is $WA+mt_1t_2t_3t_4t_5t_6 = 0$. (Observe that since $AC - B^2 = Kt_1t_2t_3t_4t_5t_6$, this means only that the equation of the sextic amphigram is of the assumed form $A\Gamma - B^2 = 0$.)

140. By what precedes we can, for a sextic amphigram which breaks up in any assigned manner, determine the value of Γ. For instance, let the amphigram break up into two cubic curves; say these are $sA + mt_1t_2t_3 = 0$, $s'A + m't_4t_5t_6 = 0$. Assume

$$A\Gamma - B^2 = \frac{mm'}{K}(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6),$$

then this equation is

$$A\Gamma - B^2 = \frac{mm'}{K}\Big[ss'A^2 + (sm't_4t_5t_6 + s'mt_1t_2t_3)A\Big] + AC - B^2;$$

that is, we have

$$\Gamma = C + \frac{mm'}{K}(ss'A + sm't_4t_5t_6 + s'mt_1t_2t_3),$$

and so in any other case.

I have already adverted to the question of the "nodal determination" of the formulae, and it might be properly here considered; viz., the question is as to the determination of the constants in such manner that, for instance, $sA + mt_1t_2t_3 = 0$ may be a nodal cubic, $UA + mt_1t_2t_3t_4 = 0$ a nodal, binodal, or trinodal quartic, &c.; but I defer it for the moment in order first to apply the theory to the quartic surfaces.

Application to Quartic Surfaces.

141. If a quartic surface has a node or nodes, we may take for a node the point $x = 0$, $y = 0$, $z = 0$; the equation of the quartic surface is then of the form

$$Aw^2 + 2Bw + \Gamma = 0,$$

where A, B, Γ are functions of x, y, z of the degrees 2, 3, 4 respectively. $A = 0$ is the tangent quadricone at the node in question; and the circumscribed cone is $A\Gamma - B^2 = 0$. By what precedes, this is an amphigram touching the quadricone along six generating lines thereof; say the tangent planes of the cone $A = 0$ along these six lines respectively are $t_1 = 0$, $t_2 = 0$, ..., $t_6 = 0$. We have then an identical equation

$$AC - B^2 = Kt_1t_2t_3t_4t_5t_6,$$

viz., regarding for a moment this equation as an equation for the determination of B, and B' as any particular solution thereof, then its general solution is $B = B' + tA$, where t is an arbitrary linear function of (x, y, z), and the B in the equation of the surface is properly $= B' + tA$. But by the substituting $w - t$ in place of w, the B of the equation of the surface would then be made $= B'$; and it thus appears that we may, without loss of generality, take the B of this equation to be any particular value B satisfying the identity in question; and then, B having such particular value, C is a quartic function of (x, y, z) completely determined by the same identity. And we then, by what precedes, at once determine Γ so that the circumscribed cone $A\Gamma - B^2 = 0$ may be a cone breaking up in any assigned manner; for instance, if it be a cone $(3, 3)$, then, as just mentioned, the two cubic cones are $sA + mt_1t_2t_3 = 0$, $s'A + m't_4t_5t_6 = 0$; and Γ has the value

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's't_4t_5t_6 + ls't_4t_5t_6),$$

above obtained.

On the Nodal Determination.

142. I am not able to discuss with much completeness the question of nodal determination. We have to consider a cubic curve $sA + mt_1t_2t_3 = 0$, a quartic curve $UA + mt_1t_2t_3t_4 = 0$, &c., as the case may be, and to determine the constants so that this shall have a node or nodes. Consider for a moment the form $PA + t_1\Omega = 0$, where Ω denotes the product $mt_2t_3\cdots$ of all or any of the tangents $t_2, \ldots, t_n$; the orders of PA, $t_1\Omega$ are of course equal, that is, the order of P is less by unity than that of Ω. I say that, by establishing a single relation between the constants, this may be made to have a node at the point of contact $A = 0$, $t_1 = 0$. In fact, writing $\Delta = \lambda\partial_x + \mu\partial_y + \nu\partial_z$, where λ, μ, ν are arbitrary, there will be a node at any point if for that point $\Delta(PA + t_1\Omega) = 0$. But for the point $A = 0$, $t_1 = 0$ this becomes $P\Delta A + \Omega\Delta t_1 = 0$; moreover, if t be any other tangent of the conic $A = 0$, and if $p = 0$ be the line joining the points of contact of the tangents t, t_1, then we may write $A = tt_1 - p^2$, and thence (since at the point in question, $A = 0$, $t_1 = 0$, we have also $p = 0$) we find $\Delta A = t\Delta t_1$, and the foregoing equation thus becomes $(tP + \Omega)\Delta t_1 = 0$; viz., this equation is satisfied irrespectively of the values of λ, μ, ν, if only at the point in question (that is, for the values of the coordinates which belong to the point $t_1 = 0$, $A = 0$) we have $tP + \Omega = 0$, which is a single relation between the constants.

143. In particular the cubic curve $sA + mt_1t_2t_3 = 0$ may be made to have a node at the point of contact of any one of the three tangents; the quartic curve $UA + mt_1t_2t_3t_4 = 0$, a node, or two or three nodes, at the point or points of contact of any one, two, or three of the four tangents; and so in other cases. These are not the only solutions, and they are in fact solutions which (as afterwards explained) I propose to reject, attending in each case only to the remaining or proper solutions of the problem.

144. To obtain in a different manner the foregoing result, consider again the cubic curve $sA+mt_1t_2t_3=0$; regarding this as a given curve, the conic $A=0$ is a conic determined (not of course completely) as a conic having therewith 3 points of 2-pointic intersection; viz., if the cubic has a node, then the cone $A=0$ is either a conic passing through the node and besides touching the curve twice, or else it is a conic touching the curve 3 times; the former is of course the above mentioned case where there is a node at one of the points of contact on the conic $A=0$; the latter is regarded as the proper solution. So in the case of a quartic curve $UA+mt_1t_2t_3t_4=0$, regarding this as a given curve, the conic $A=0$ is a conic having therewith four points of 2-pointic intersection; viz., if the quartic curve has one, two, or three nodes, then the conic is either a conic passing through one, two, or three nodes, and besides touching the quartic thrice, twice, or once; or else it is a conic touching the quartic four times. The former is the above mentioned case where there is a node or nodes at a point or points of contact with the conic $A=0$; the latter is regarded as the proper solution.

145. To fix the ideas, and at the same time obtain a result which will be afterwards useful, I work out the formulae for the cubic curve $sA+mt_1t_2t_3=0$, taking this equation under the form

$$s\left(\tfrac{x}{\lambda}+\tfrac{y}{\mu}+\tfrac{z}{\nu}\right)+\tfrac{m}{\lambda\mu\nu}\left(x^2+y^2+z^2-2yz-2zx-2xy\right)+\tfrac{m}{\lambda\mu\nu}\,xyz=0.$$

This may have a node in two different ways; viz.,

1°. At the point of contact of one of the tangents $x=0$, $y=0$, $z=0$ with the conic $A=0$; say at the point of contact of $x=0$, that is, the point $x=0$, $y=z=0$. The value of m is $\dfrac{4}{\mu}+\dfrac{4}{\nu}$; hence $\dfrac{m}{s}=\dfrac{4}{\mu}+\dfrac{4}{\nu}$; and, substituting, the equation of the curve becomes

$$\tfrac{s\,x}{\lambda}\left[\left(\tfrac{x}{\lambda}\right)+\tfrac{1}{\mu\nu}\left\{x^2-2x(y+z)+(y-z)^2\right\}\right]+\tfrac{1}{\lambda\mu\nu}\,m\,(x+y-z)^2=0,$$

which has obviously a node at the point in question.

2°. The node may be at a point not on the conic $A=0$, viz. the value of $\dfrac{m}{s}$ is $-\dfrac{4\lambda}{\mu\nu}$; the equation is

$$s\left(\tfrac{x}{\lambda}+\tfrac{y}{\mu}+\tfrac{z}{\nu}\right)-\tfrac{m}{\lambda\mu\nu}\left(x^2+y^2+z^2-2yz-2zx-2xy\right)+\tfrac{m}{\lambda\mu\nu}\,xyz=0.$$

In fact, writing for shortness

$$\begin{aligned}-\lambda+\mu+\nu&=L,\\ \lambda-\mu+\nu&=M,\\ \lambda+\mu-\nu&=N,\\ \lambda+\mu+\nu&=P,\end{aligned}$$

the node is at the point $x:y:z=L\lambda:M\mu:N\nu$; which is at once verified, if we remark that, writing for convenience $x,y,z=L\lambda,M\mu,N\nu$, then we have

$$\begin{aligned}-x+y+z&=MN,\\ x-y+z&=NL,\\ x+y-z&=LM,\\ \tfrac{x}{\lambda}+\tfrac{y}{\mu}+\tfrac{z}{\nu}&=P,\quad x^2+y^2+z^2-2yz-2zx-2xy=-LMNP\ (=A).\end{aligned}$$

For, of the three equations for the coordinates of a node, the first is

$$-\frac{s}{\lambda}LMNP+P(-2MN)+\frac{m}{\lambda\mu\nu}\,MN\mu\nu=0,$$

that is, for the values in question,

$$-LMNP-2\lambda MNP+P^2MN=0,$$

or, finally, $-P-2\lambda+P=0$, which is satisfied; and similarly the other two equations are satisfied.

Quartic Surfaces resumed.

146. Passing now from curves to cones, and to the theory of the quartic surface, suppose that there is a component cone having a nodal line, say the cubic cone $sA + mt_1t_2t_3 = 0$: if the remaining factor is $t_4t_5t_6$, then we have

$$\Gamma = C + \frac{m}{K}\, s\, t_4t_5t_6.$$

Suppose the nodal line is a line of contact with the cone $A = 0$, say its equations are $t_1 = 0$, $p = 0$ (p a linear function), then $sA + mt_1t_2t_3$ is a quadric function $(*\,|\,t_1, p)^2$ (of course with variable coefficients); hence $A\Gamma - B^2$ is a quadric function; and B being a linear function $(*\,|\,t_1, p)$, it follows that Γ is a linear function, and thence that Γ is also a linear function; that is, A, B, Γ are each of them a linear function $(*\,|\,t_1, p)$, or the line in question (viz. the line of contact $A = 0$, $t_1 = 0$) is a line on the quartic surface $Aw^2 + 2Bw + \Gamma = 0$. As already mentioned, *I exclude from consideration the surfaces which have upon them a line through two nodes*; that is, I exclude from consideration the case in question where any component cone, or say where the sextic cone, has a nodal line which is a line on the tangent cone $A = 0$.

147. Now, excluding the case just referred to, *I assume as a postulate* that there is but one way in which the cubic cone $sA + mt_1t_2t_3 = 0$ can be made to have a nodal line, or the quartic cone $UA + mt_1t_2t_3t_4 = 0$ one, two, or three nodal lines &c., as the case may be. It is to be understood that this does not mean that the constants are in any of these cases completely determined, but that there is between them a relation or relations constituting a general solution which includes in itself every particular solution whatever. I have no doubt that as regards the cubic cone at least the assumption is correct. This being so, the character of a single node determines the nature of the surface; for instance, if there is a node $(3_1, 3)$, then taking this as the point $(x = 0, y = 0, z = 0)$ the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, where

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's\, t_4t_5t_6 + ls't_1t_2t_3),$$

a surface of a determinate nature; so that the character of all the remaining nodes is completely determined.

148. The point to be attended to is, that if for instance there were two essentially distinct ways of giving the cubic cone $sA + mt_1t_2t_3 = 0$ a nodal line (such as there would be if the excluded case were considered admissible), then the foregoing equation of the surface would or might include two distinct forms of equation applying to different kinds of surface. The conclusion is that there is but one kind of quartic surface having a node $(3_1, 3)$. Admitting this, and similarly that there is but one kind of quartic surface having a node 6_{10}, it follows that if (as the fact is) there is a surface having the nodes $1(6_{10}) + 10(3_1, 3)$ (Kummer's 11-nodal surface), then that the two first-mentioned kinds are in fact each of them this last-mentioned kind of surface; and it was in this manner that I arrived at the enumeration given near the beginning of the present Memoir.

149. The reasoning is, of course, in place of a direct demonstration which would consist in showing that a surface having a node $(3_1, 3)$ has 9 other like nodes, and also a node 6_{10}; and that a surface having a node 6_{10} has 10 other nodes $(3_1, 3)$; and that, starting from either form of equation, we could, by passing to a node of the other kind, obtain the other form of equation.

Enumeration of the Cases.

150. I collect the results as follows: I call to mind that we have always the identical equation $AC - B^2 = Kt_1t_2t_3t_4t_5t_6$, that the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, and that the circumscribed cone is $A\Gamma - B^2 = 0$. The equation of a surface having different kinds of nodes will assume different forms according as the origin (or point $x = 0$, $y = 0$, $z = 0$) is taken to be at a node of one or other of these kinds; these forms of the equations are distinguished as "node-forms," viz., we speak of the node-form $(3_1, 3)$ when the origin is a node $(3_1, 3)$, and so in other cases.

The 16-nodal surface $16(1,1,1,1,1,1)$.
node-form $(1,1,1,1,1,1)$,
cone is $t_1t_2t_3t_4t_5t_6 = 0$,
and $\Gamma = C$;
viz., equation is $Aw^2 + 2Bw + C = 0$.

The 15-nodal surface $15(2,1,1,1,1)$.
node-form $(2,1,1,1,1)$,
cone is $(lA + mt_1t_2)\,t_3t_4t_5t_6 = 0$,
and $\Gamma = C + \dfrac{Kl}{m}\,t_3t_4t_5t_6$.

The 14-nodal surface $8(3_1,1,1,1) + 6(2,2,1,1)$.
node-form $(3_1,1,1,1)$,
cone is $(sA + mt_1t_2t_3)\,t_4t_5t_6 = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{Ks}{m}\,t_4t_5t_6$.

node-form $(2,2,1,1)$,
cone is $(lA + mt_1t_2)(l'A + m't_3t_4)\,t_5t_6 = 0$,
and $\Gamma = C + \dfrac{mm'}{K}(ll'A + lm't_3t_4 + l'mt_1t_2)\,t_5t_6$.

The 13(α)-nodal surface $3(4_1,1,1) + 1(3,1,1,1) + 9(3_1,2,1)$.
node-form $(4_1,1,1)$,
cone is $(UA + mt_1t_2t_3t_4)\,t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a trinodal quartic 4_1,
and $\Gamma = C + \dfrac{KU}{m}\,t_5t_6$.

node-form $(3,1,1,1)$,
cone is $(sA + mt_1t_2t_3)\,t_4t_5t_6 = 0$,
and $\Gamma = C + \dfrac{Ks}{m}\,t_4t_5t_6$.

node-form $(3_1,2,1)$,
cone is $(sA + mt_1t_2t_3)(l'A + m't_4t_5)\,t_6 = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(sl'A + m's\,t_4t_5 + ml't_1t_2t_3)\,t_6$.

The 13(β)-nodal surface $13(2,2,2)$.
node-form $(2,2,2)$,
cone is $(lA + mt_1t_2)(l'A + m't_3t_4)(l''A + m''t_5t_6) = 0$,
and

$$\begin{aligned}\Gamma = C + \frac{mm'm''}{K^2}\big[&ll'l''A^2 + (l'l''mt_1t_2 + l''lm't_3t_4 + ll'm''t_5t_6)A\\ &+lm'm''t_3t_4t_5t_6 + l'm''m\,t_5t_6t_1t_2 + l''mm'\,t_1t_2t_3t_4\big].\end{aligned}$$

The 12(α)-nodal surface $12(4_1,2)$.
node-form $(4_2,2)$,
cone is $(UA + mt_1t_2t_3t_4)(l'A + m't_5t_6) = 0$,

where $UA + mt_1t_2t_3t_4 = 0$ is a trinodal quartic 4_3,
and $\Gamma = C + \dfrac{mm'}{K}(l'UA + m'Ut_5t_6 + l'm\, t_1t_2t_3t_4)$.

The $12(\beta)$**-nodal surface** $\quad 2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$**.**
node-form $(5_1, 1)$,
cone is $(VA + mt_1t_2t_3t_4t_5)\, t_6 = 0$,
where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 6-nodal quintic 5_1,
and $\Gamma = C + \dfrac{KV}{m}\, t_6$.

node-form $(3_1, 3_1)$,
cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,
where $sA + mt_1t_2t_3 = 0$ and $s'A + m't_4t_5t_6 = 0$ are each of them a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(ss'A + m's\, t_4t_5t_6 + ms't_1t_2t_3)$.

node-form $(3, 2, 1)$,
cone is $(sA + mt_1t_2t_3)(l'A + m't_4t_5)\, t_6 = 0$,
and $\Gamma = C + \dfrac{mm'}{K}(sl'A + m's\, t_4t_5 + l'm\, t_1t_2t_3)\, t_6$.

The $12(\gamma)$**-nodal surface,** $\quad 12(4_1, 1, 1)$**.**
node-form $(4_1, 1, 1)$,
cone is $(UA + mt_1t_2t_3t_4)\, t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_3,
and $\Gamma = C + \dfrac{KU}{m}\, t_5t_6$.

The $11(\alpha)$**-nodal surface,** $\quad 1(6_{10}) + 10(3_1, 3)$**.**
node-form (6_{10}),
cone is $WA + mt_1t_2t_3t_4t_5t_6 = 0$,
where this is a 10-nodal sextic 6_{10},
and $\Gamma = C + \dfrac{KW}{m}$.

node-form $(3_1, 3)$,
cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,
where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1,
and $\Gamma = C + \dfrac{mm'}{K}(ss'A + m's\, t_4t_5t_6 + ms't_1t_2t_3)$.

Other 11**-nodal surfaces.**
node-form $(5_1, 1)$,
cone is $(VA + mt_1t_2t_3t_4t_5)\, t_6 = 0$,
where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 5-nodal quintic 5_1,
and $\Gamma = C + \dfrac{KV}{m}\, t_6 = 0$;

node-form $(4_2, 2)$,
cone is $(UA + mt_1t_2t_3t_4)(l'A + m't_5t_6) = 0$,
where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_2,
and $\Gamma = C + \dfrac{mm'}{K}(l'UA + m'Ut_5t_6 + l'm\, t_1t_2t_3t_4)$;

node-form $(4, 1, 1)$,
cone is $(UA + mt_1t_2t_3t_4)\, t_5t_6 = 0$,
where $UA + mt_1t_2t_3t_4$ is a nodal quartic 4_1,
and $\Gamma = C + \dfrac{KU}{m}\, t_5t_6$,
but whether these node-forms belong to the same or to different surfaces is not ascertained.

The enumeration is not extended to the 10-nodal surfaces, but I consider one case of these surfaces.

The $10(\alpha)$**-nodal surface** $10(3,3)$**.**

151. I assume only that there is a single node $(3,3)$: taking the cone to be

$$(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0;$$

then for the equation of the surface, in the node-form $(3,3)$ in question, we have

$$\Gamma = C + \frac{mm'}{K}(ss'A + sm't_4t_5t_6 + s'm\,t_1t_2t_3).$$

But I present this result under a different form, as follows: I write

$$A = p^2 + ft_1t_2 = q^2 + gt_3t_4 = r^2 + ht_5t_6;$$

where f, g, h are constants, and, as before, $p=0$, $q=0$, $r=0$ are the lines joining the points of contact of t_1, t_2; t_3, t_4; and t_5, t_6 respectively: we have

$$sA + mt_1t_2t_3 = sA + mt_3\Big(\frac{A-q^2}{g}\Big), \quad \text{and} \quad s'A + m't_4t_5t_6 = s'A\Big(\cdots\Big),$$

or in place of s, s' introducing new linear functions σ, σ', the cubic curves may be taken to be $\sigma A - \dfrac{m}{g}\,q^2t_3$, $\sigma'A - \dfrac{m'}{h}\,r^2t_5$, so that we have

$$\frac{mm'}{K}\Big(\sigma A - \tfrac{m}{g}\,q^2t_3\Big)\Big(\sigma'A - \tfrac{m'}{h}\,r^2t_5\Big) + \cdots = A\Big(\cdots\Big) - \frac{mm'}{fgh}\Big(\cdots\Big);$$

whence $B = \dfrac{mm'}{fgh}\,(pqr + tA)$, where t is a linear function of the coordinates; and we then have

$$\Gamma = C + \frac{mm'}{K}\big[\sigma\sigma'A + \cdots\big],$$

where A may be considered as standing for $q^2 + gt_3t_4$. The equation $Aw^2 + 2Bw + \Gamma = 0$ of the surface, substituting throughout for A its value, is therefore

$$\Big(\tfrac{1}{g}\Big)(q^2 + gt_3t_4)\,w^2 + 2\big\{pqr + t(q^2 + gt_3t_4)\big\}\,w + (\cdots) = 0,$$

where the cone is $\cdots = 0$.

152. Writing in the equation of the surface $w\sqrt{f}$ instead of w, it becomes

$$(q^2 + gt_3t_4)\,w^2 + 2\big[pqr + t(q^2 + gt_3t_4)\big]\,w + \tfrac{fgh}{mm'}\Big[\tfrac{m'}{f}(q^2 + gt_3t_4) + 2tpqr\Big] = 0,$$

and then writing $\dfrac{m}{f}\sigma$ and $\dfrac{m'}{h}\sigma'$ for σ and σ' respectively, this is

$$(q^2 + gt_3t_4)\,w^2 + 2pqrw + 2tw(q^2 + gt_3t_4) + g\big[\cdots\big] + t^2(q^2 + gt_3t_4) + 2tpqr = 0.$$

We may consider t_3, t_4 as denoting not the functions originally so represented, but these functions each multiplied by a suitable constant, and thereupon write $g = -1$; viz., $t_3 = 0$, $t_4 = 0$, will now denote any two tangents to the conic $A = 0$, the implicit factors being so determined that $A = q^2 - t_3t_4$. The equation of the surface is

$$(q^2 - t_3t_4)\,w^2 + 2pqrw + 2tw(q^2 - t_3t_4)$$
$$-\sigma\sigma'(q^2 - t_3t_4) + \sigma r^2t_5 + \sigma' p^2t_3 + p^2r^2 + t^2(q^2 - t_3t_4) + 2tpqr = 0;$$

viz., this is

$$-(q^2 - t_3t_4)\big[(w+t)^2 - \sigma\sigma'\big] + 2pqr(w+t) + \sigma r^2t_5 + \sigma' p^2t_3 + p^2r^2 = 0,$$

the sextic cone being

$$-\big[\sigma(q^2 - t_3t_4) - p^2t_3\big]\cdot\big[\sigma'(q^2 - t_3t_4) - r^2t_5\big] = 0.$$

153. But the foregoing equation of the surface is

$$\begin{vmatrix} -\sigma', & w+t, & \cdot, & r \\ w+t, & -\sigma, & p, & \cdot \\ \cdot, & p, & -q, & t_3 \\ r, & \cdot, & t_3, & -q \end{vmatrix} = 0,$$

as is at once seen by developing the determinant; the functions $w+t$, σ, σ', p, q, r, t_3, t_5 are all of them linear; and the determinant is thus a symmetrical quartic determinant the terms whereof are linear functions of the coordinates; viz. the surface is a symmetroid. That is, a surface having a single node $(3,3)$ is a symmetroid; but I have shown (Second Memoir, No. 116) that a symmetroid has each of its ten nodes $(3,3)$; wherefore the surface having a single node $(3,3)$ is the $10(\alpha)$-nodal surface, nodes $10(3,3)$.

154. Start from two cubic cones $U = 0$, $V = 0$, having each the same vertex $(x = 0, y = 0, z = 0)$; we may in a variety of ways determine the two cones $\alpha U + \beta V = 0$, $\gamma U + \delta V = 0$, having a common inscribed quadric cone $A = 0$ (viz., $\alpha : \beta$ being assumed at pleasure, then $\gamma : \delta$ will be determined; not, I believe, uniquely, but I do not know what the multiplicity is). This being so, the quadric cone $A = 0$ is uniquely determined; and then, assuming at pleasure the plane $w = 0$, the $10(\alpha)$-nodal surface $Aw^2 + 2Bw + \Gamma = 0$ is uniquely determined: consequently the remaining nine nodes are determinate points on the nine lines $U = 0$, $V = 0$ respectively. And we have thus a system of ten points in space such that, joining any one of them with the remaining nine, the nine lines so obtained are the intersections of two cubic cones, or say that they are an ennead of lines.

Notation for the Cases afterwards considered.

155. I proceed to further develop the theory of some of the different surfaces. The same node-form of equation will, of course, assume different shapes according to the actual expressions in terms of the coordinates (x, y, z) of the several functions A, B, &c., which enter into it. I have found it convenient to attribute to A and B certain specific values which are not in every case those of the coefficients of w^2, w in the equation of the surface: this means that we must, in the equation of the surface, substitute new symbols for these coefficients, and write the equation say in the form $A'w^2 + 2B'w + \Gamma' = 0$; the change of notation, when it occurs, will be duly explained.

156. It is in general (but not always) convenient to take the equation of the tangent cone to be $x^2+y^2+z^2-2yz-2zx-2xy=0$; for then any plane $\dfrac{x}{\alpha}+\dfrac{y}{\beta}+\dfrac{z}{\gamma}=0$, where $\alpha+\beta+\gamma=0$, will be a tangent plane; so that six tangent planes may be represented by $x=0$, $y=0$, $z=0$, and by three equations of the form just referred to. And in reference to this assumed form of the equation of the tangent cone, and to what follows, I write

$$\begin{aligned}
&\alpha+\beta+\gamma=0,\\
&\alpha'+\beta'+\gamma'=0,\\
&\alpha''+\beta''+\gamma''=0,\\
&P\ =\tfrac{x}{\alpha}+\tfrac{y}{\beta}+\tfrac{z}{\gamma},\\
&P'=\tfrac{x}{\alpha'}+\tfrac{y}{\beta'}+\tfrac{z}{\gamma'},\\
&P''=\tfrac{x}{\alpha''}+\tfrac{y}{\beta''}+\tfrac{z}{\gamma''},
\end{aligned}$$

$$\begin{aligned}
X&=\alpha\,(\gamma'\gamma''y-\beta'\beta''z),\\
Y&=\beta\,(\alpha'\alpha''z-\gamma'\gamma''x),\\
Z&=\gamma\,(\beta'\beta''x-\alpha'\alpha''y),\\
X'&=\alpha'\,(\gamma''\gamma y-\beta''\beta z),\\
Y'&=\beta'\,(\alpha''\alpha z-\gamma''\gamma x),\\
Z'&=\gamma'\,(\beta''\beta x-\alpha''\alpha y),\\
X''&=\alpha''\,(\gamma\gamma' y-\beta\beta' z),\\
Y''&=\beta''\,(\alpha\alpha' z-\gamma\gamma' x),\\
Z''&=\gamma''\,(\beta\beta' x-\alpha\alpha' y),
\end{aligned}$$

$$\begin{aligned}
A&=x^2+y^2+z^2-2yz-2zx-2xy,\\
B&=\alpha\alpha'\alpha''\,(y^2z-yz^2)+\beta\beta'\beta''\,(z^2x-zx^2)+\gamma\gamma'\gamma''\,(x^2y-xy^2)+M\,xyz,\\
C&=(\alpha\alpha'\alpha''yz+\beta\beta'\beta''zx+\gamma\gamma'\gamma''xy)^2-\tfrac{1}{4}\cdots
\end{aligned}$$

where

$$\begin{aligned}
M&=(\beta-\gamma)\alpha'\alpha''+(\gamma-\alpha)\beta'\beta''+(\alpha-\beta)\gamma'\gamma'',\\
&=(\beta'-\gamma')\alpha''\alpha+(\gamma'-\alpha')\beta''\beta+(\alpha'-\beta')\gamma''\gamma,\\
&=(\beta''-\gamma'')\alpha\alpha'+(\gamma''-\alpha'')\beta\beta'+(\alpha''-\beta'')\gamma\gamma',
\end{aligned}$$

$$\tfrac{1}{2}\cdots=\big[(\beta-\gamma)(\beta'-\gamma')(\beta''-\gamma'')+(\gamma-\alpha)(\gamma'-\alpha')(\gamma''-\alpha'')+(\alpha-\beta)(\alpha'-\beta')(\alpha''-\beta'')\big];$$

also

$$A^2=4\alpha\alpha'\alpha''\beta\beta'\beta''\gamma\gamma'\gamma'';$$

and we have identically [an identity follows from this notation].

The 16-nodal Surface $16(1,1,1,1,1,1)$.

157. Kummer starts from an irrational equation, which is readily converted into the following

$$\sqrt{x(X-w)}+\sqrt{y(Y-w)}+\sqrt{z(Z-w)}=0,$$

and then, rationalizing, we have

$$Aw^2+2Bw+C=0,$$

where as above

$$AC-B^2=K\,xyz\,PP'P''.$$

This agrees with the foregoing theory; viz., the point $(x = 0, y = 0, z = 0)$ being a node, the rationalized equation must, of course, be in the node-form $(1,1,1,1,1,1)$, (being the only node-form); and the symmetry of the formulae enables us at once to write down the equations of the 16 singular planes, and thence to deduce the coordinates of the 16 nodes; viz.,

the singular planes are	**and the nodes are**
(1) $x = 0$,	(1) $(0,\ -\beta,\ \gamma,\ \alpha'\alpha''\beta\gamma)$,
(2) $y = 0$,	(2) $(\alpha,\ \cdot,\ -\gamma,\ \beta'\beta''\gamma\alpha)$,
(3) $z = 0$,	(3) $(-\alpha,\ \beta,\ \cdot,\ \gamma'\gamma''\alpha\beta)$,
(4) $w = 0$,	(4) $(\alpha'\alpha'',\ \beta'\beta'',\ \gamma'\gamma'',\ \cdot)$,
(5) $X - w = 0$,	(5) $(1,\ \cdot,\ \cdot,\ \cdot)$,
(6) $Y - w = 0$,	(6) $(\cdot,\ 1,\ \cdot,\ \cdot)$,
(7) $Z - w = 0$,	(7) $(\cdot,\ \cdot,\ 1,\ \cdot)$,
(8) $P = 0$,	(8) $(\cdot,\ \cdot,\ \cdot,\ 1)$,
(9) $X' - w = 0$,	(9) $(\cdot,\ -\beta',\ \gamma',\ \alpha''\alpha\beta'\gamma')$,
(10) $Y' - w = 0$,	(10) $(\alpha',\ \cdot,\ -\gamma',\ \beta''\beta\gamma'\alpha')$,
(11) $Z' - w = 0$,	(11) $(-\alpha',\ \beta',\ \cdot,\ \gamma''\gamma\alpha'\beta')$,
(12) $P' = 0$,	(12) $(\alpha''\alpha,\ \beta''\beta,\ \gamma''\gamma,\ \cdot)$,
(13) $X'' - w = 0$,	(13) $(\cdot,\ -\beta'',\ \gamma'',\ \alpha\alpha'\beta''\gamma'')$,
(14) $Y'' - w = 0$,	(14) $(\alpha'',\ \cdot,\ -\gamma'',\ \beta\beta'\gamma''\alpha'')$,
(15) $Z'' - w = 0$,	(15) $(-\alpha'',\ \beta'',\ \cdot,\ \gamma\gamma'\alpha''\beta'')$,
(16) $P'' = 0$,	(16) $(\alpha\alpha',\ \beta\beta',\ \gamma\gamma',\ \cdot)$,

where the nodes and planes are numbered as by Kummer; and by means of his (differently arranged) diagram of the relation between the several nodes and planes, I was enabled to form the following square diagram, which exhibits this relation in, I think, the most convenient form. To explain this, observe that in the upper and left-hand margins, the numbers refer to the nodes; in the body of the table, and in the right-hand margin to the planes, the table shows that for the node 1, the circumscribed cone is made up of the planes $1, 6, 7, 8, 9, 13$; and that the remaining 15 nodes are situate on the nodal lines of this cone, the node 2 on the intersection of the planes $7, 8$; the node 3 on the intersection of the planes $6, 8$, and so on; and the like as regards the other lines of the table.

[The original page 283 here gives a 16×16 square diagram (Table No. 12) exhibiting the incidence of the 16 nodes with the 16 singular planes of the 16-nodal surface. Each row i shows the planes through node i (the six planes whose intersection is the node), and the body of the table records, in row i column j, the pair of planes whose intersection at node j lies on the sextic cone of node i. The right-hand margin lists the planes meeting at the node of that row: $1 \ldots\ 1, 6, 7, 8, 9, 13$; $2 \ldots\ 2, 6, 8, 10, 14, \ldots$; etc. (See PDF page 283 of the volume for the full table.)]

158. The before mentioned irrational equation may be written

$$\sqrt{1 \cdot 5} + \sqrt{2 \cdot 6} + \sqrt{3 \cdot 7} = 0,$$

and by symmetry we see that also

$$\sqrt{1 \cdot 9} + \sqrt{2 \cdot 10} + \sqrt{3 \cdot 11} = 0,$$

$$\sqrt{1 \cdot 13} + \sqrt{2 \cdot 14} + \sqrt{3 \cdot 15} = 0;$$

viz., these are three equations each containing the planes 1, 2, 3, which are three of the planes belonging to the node 1; the other three planes in any such equation (for instance, the planes 5, 6, 7, in the first equation) being three planes belonging to another node. Instead of the planes 1, 2, 3, we may have any other three planes belonging to the node 1; and instead of the node 1, any other node; but each equation belongs to two nodes: the number of equations is thus

$$\tfrac{6\cdot5\cdot4}{1\cdot2\cdot3} \times 16 \times 3 \div 2, \; = 480.$$

159. To obtain the planes belonging to any such equation, combine any two of the outside right-hand lines of the diagram, these contain in common two numbers the places of which are interchanged; striking these out, we have four columns, and taking out of these any three columns, we have the corresponding sets of planes. For instance, lines 1 and 2 contain 7 8 and 8 7 respectively; striking these out, the lines are

$$1, \quad 9, 13, \quad 6;$$

$$2, \quad 10, 14, \quad 5;$$

whence we have the sets $(1, 9, 13)$ and $(2, 10, 14)$; viz., there is an irrational equation of the form

$$\sqrt{1\cdot2} + \sqrt{9\cdot10} + \sqrt{13\cdot14} = 0,$$

but it is probably necessary to introduce constant factors along with the products $1\cdot2$, $9\cdot10$, and $13\cdot14$ respectively. There are $\frac{1}{2}16\cdot15$, $= 120$ pairs of lines, and each line gives 4 equations; in all 120×4, $= 480$ equations, as above.

160. I stop to remark that Kummer gives for his 13-nodal surface an equation containing three arbitrary constants, say λ, μ, ν, such that, putting one of these $= 0$, we have the 14-nodal surface; putting two of them each $= 0$, the 15-nodal surface; putting all three of them $= 0$, the 16-nodal surface. The equations for the 16-nodal surface and that for the 14-nodal surface, made use of by Kummer, are, in fact, those deduced as above from the equation of the $13(\alpha)$-nodal surface; and the like form might have been used for the 15-nodal surface. But the form actually used by Kummer, as presently appearing, is an equivalent form not thus deducible from the equation of the $13(\alpha)$-nodal surface.

The 15-nodal Surface $15(2, 1, 1, 1, 1)$.

161. Kummer's equation is readily converted into the following:

$$Aw^2 + 2Bw + C + \tfrac{Kl}{m}\, xyzP = 0,$$

the circumscribed cone being thus

$$(lA + mP'P'')\, xyzP = 0,$$

and the equation being in the node-form $(2, 1, 1, 1, 1)$.

The formulae for the 15 nodes and the 10 singular planes depend upon a quadric equation, for the symmetrical expression of which I write

$$\omega = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma''\alpha' = \gamma'\gamma'' - \alpha''\beta',$$

$$\varpi = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma'\alpha'' = \gamma'\gamma'' - \alpha'\beta'';$$

so that

$$\omega - \varpi = \frac{\beta'\gamma'' - \beta''\gamma'}{\alpha} = \frac{\gamma'\alpha'' - \gamma''\alpha'}{\beta} = \frac{\alpha'\beta'' - \alpha''\beta'}{\gamma},$$

$$\omega + \varpi = \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'' + \cdots;$$

the equation in question then is

$$(\rho - \omega)(\rho - \varpi) - \frac{Kl}{4\alpha\beta\gamma m} = 0;$$

so that, calling the roots of it ρ_1, ρ_2, we have

$$\rho_1 + \rho_2 = \omega + \varpi, \quad \rho_1\rho_2 = \omega\varpi + \frac{Kl}{4\alpha\beta\gamma m};$$

or we may write

$$\rho_1 = \tfrac{1}{2}(\omega + \varpi + \sqrt{\Omega}), \quad \rho_2 = \tfrac{1}{2}(\omega + \varpi - \sqrt{\Omega}),$$

if for shortness

$$\Omega = (\omega - \varpi)^2 - \frac{Kl}{\alpha\beta\gamma m}.$$

162. I write also for shortness

$$\begin{aligned}
\mathfrak{a} &= (\beta - \gamma)\alpha'\alpha'' + \alpha(\beta'\beta'' - \gamma'\gamma''),\\
\mathfrak{b} &= (\gamma - \alpha)\beta'\beta'' + \beta(\gamma'\gamma'' - \alpha'\alpha''),\\
\mathfrak{c} &= (\alpha - \beta)\gamma'\gamma'' + \gamma(\alpha'\alpha'' - \beta'\beta'');
\end{aligned}$$

and I say that the singular planes are

$$\begin{aligned}
&(1)\ x = 0, \quad (2)\ y = 0, \quad (3)\ z = 0,\\
&(9)\ \ -\tfrac{\gamma'\beta'}{\alpha}\,p_1(Z' - w) + (\rho_1 - \omega)\left(\tfrac{x}{\beta} + \tfrac{y}{\gamma}\right) = 0,\\
&(10)\ \ -\tfrac{\beta'\alpha'}{\gamma}\,p_1(Y' - w) + (\rho_1 - \omega)\left(\tfrac{z}{\alpha} + \tfrac{x}{\beta}\right) = 0,\\
&\quad \cdots\\
&P = 0;
\end{aligned}$$

[The full lists of 15 nodes and 10 singular planes, with their pairings to Kummer's numbering, occupy book pages 285–286. The formulae are routine consequences of the symmetric functions $\rho_1, \rho_2, \mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ above. See the PDF pages 307–308 of the scan for the literal expressions.]

163. The small reference numbers are those used by Kummer. It is, I think, better to retain the reference numbers belonging to the case of the 16-nodal surface; viz., there are here given, large, $1, 2, 3, 8, 9, 10, 11, 13, 14, 15$ for the planes, and $1, 2, 3, 5, 6, \ldots$ to 16 for the nodes. Belonging to each node (that is, with the node as vertex) there is a quadric cone passing through 8 other nodes; and each node lies (exclusively of the cone whose vertex it is) in 8 such cones. We have thus the following square diagram:

[The original page 287 here gives a 15×15 square diagram exhibiting, for the 15-nodal surface, the incidence of nodes with planes and quadric cones C, where the cone C replaces a pair of the planes of the 16-nodal table. See PDF page 287 for the table.]

164. The arrangement is the same as in the 16-nodal square diagram; only, in the right-hand margin, a bracket $(6, 7)$ denotes that instead of the planes $(6, 7)$ we have a quadric cone; which cones are, in the body of the table, denoted by C. Thus for the node 1 the sextic cone is made up of the planes 1, 8, 9, 13 and of a quadric cone $(6, 7)$, $= C$: the remaining 14 nodes lie on the nodal lines of the sextic cone, viz., the node 2 on an intersection of the cone C with the plane 8, the node 3 on an intersection of the same cone and plane, the node 5 on the intersection of the planes 9, 13, and so on.

The Equation of the 15-nodal Surface, as deduced from that of the 13(α)-nodal.

165. If, in the equation hereafter given for the 13-nodal surface, we write $\nu = 0$, $\mu = 0$, or (what is the same thing) in that of the 14-nodal surface we write $\mu = 0$, the form is

$$\big[A + 4\lambda yz\big]\, w^2 + 2w\big[B - 2\lambda yzX\big] + C = 0.$$

The circumscribed sextic cone is thus $(2, 1, 1, 1, 1)$,

$$(\alpha\alpha''\beta\beta''yz + \beta\gamma\alpha''zx + \gamma\alpha\beta''xy + \alpha\beta xz)\, yz\, (P' - p_1P'')(P' - p_2P'') = 0,$$

where p_1, p_2 now denote the roots of the equation

$$PP'\,(p\alpha - \alpha''\gamma' - \alpha'\gamma'')\,(\cdots) + (\cdots)\, p\alpha'\alpha'' = 0.$$

The singular planes are

$$\begin{aligned}
&w = 0,\\
&P' - p_1P'' = 0,\\
&P' - p_2P'' = 0,\\
&z = 0,\\
&y = 0,\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - (Z'' - w) = 0,\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - (Z'' - w) = 0,\\
&\tfrac{\beta'\gamma''}{\alpha'\alpha''}\, p_2\,(Y' - w) - (Y'' - w) = 0,\\
&\tfrac{\beta'\gamma''}{\alpha'\alpha''}\, p_1\,(Y' - w) - (Y'' - w) = 0,\\
&Z - w = 0,
\end{aligned}$$

and we have then the same square table as before: the coordinates of the 15 nodes may be obtained without difficulty.

166. The form is really equivalent to that first considered in regard to the 15-nodal surface. To show that this is so, we have only to arrange according to powers of x; viz., the equation thus becomes

$$\begin{aligned}
&x^2\big[(\gamma'\gamma'')^2y^2 + (\beta'\beta'')^2z^2 + w^2 - 2w\beta'\beta''z + 2w\gamma'\gamma''y + 2\beta'\beta''\gamma'\gamma''yz\big]\\
&\ + 2x\big[\cdots\big] + \big[aa'\alpha''yz + wy - wz\big]^2 - 4\lambda yzw(X - w) = 0,
\end{aligned}$$

where, if for a moment A denotes the coefficient of x^2, we have $y = 0$, $z = 0$, $w = 0$, $X - w = 0$, four tangent planes of the quadric cone $A = 0$.

14-nodal Surface $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$, Node-form $(3_1, 1, 1, 1)$.

167. In the equation hereafter given for the 13-nodal surface, writing $\nu = 0$, the sextic cone becomes

$$4z(\lambda\alpha y^2 + \mu\beta zx + \beta'\gamma''xy + \gamma\alpha xy^2 + \alpha\beta xyz)\,(P' - p_1P'')(P' - p_2P'') = 0;$$

viz., this is of the form in question $(3_1, 1, 1, 1)$; and the equation of the surface is

$$w^2\big\{A + 4(\lambda y + \mu x)z - 4\lambda\mu z^2\big\} + 2w\big\{B - 2(\lambda yzX + \mu zxY)\big\} + C = 0.$$

The singular planes are

$$\begin{aligned}
&w = 0, && (4)\\
&P' - p_1 P'' = 0, && (12)\\
&P' - p_2 P'' = 0, && (16)\\
&z = 0, && (3)\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - (Z'' - w) = 0, && (15)\\
&\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - (Z'' - w) = 0, && (11)
\end{aligned}$$

and the nodes are

$$(0,0,0,1),\quad (1,0,0,0),\quad (0,1,0,0),\quad (0,0,1,0),$$

together with the 11 other nodes corresponding (with Kummer's small numbers in parentheses) to the table on page 289 of the scan.

168. In these formulae, p_1, p_2 denote the roots of the equation

$$\begin{aligned}
&\big[\lambda(\beta'\gamma'')^2 + \mu(\alpha\beta'')^2\big](p\alpha\beta\gamma)^2\\
&\quad + \big[\lambda(\beta'\gamma')^2 + \mu(\gamma'\alpha')^2\big](\alpha''\beta'\gamma'')^2\\
&\quad - 2\big[\lambda\beta'\beta'' + \mu\alpha'\alpha'' + 2\gamma'\gamma''(\omega - \varpi)^2\big]\, p\alpha\beta\gamma\,\alpha''\beta'\gamma'' = 0.
\end{aligned}$$

169. The relation of the nodes and sextic cones is given by means of the square diagram on the opposite page:

[The original page 291 here gives a 14×14 square diagram of the 14-nodal surface, with notations C for the nodal cubic cone (a cubic replacing planes $5, 6, 8$), D, D' for two quadric cones (replacing the planes $2, 5$ and $9, 13$ respectively), and $\times$ denoting the nodal line of the cubic. Vacant squares are understood as (D, D'). See PDF page 291 for the full table.]

170. The equation of the surface may be presented in the irrational form

$$\begin{aligned}
&\sqrt{(P' - p_1P'')}\big\{\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_2\,(Z' - w) - \gamma'\alpha''\beta''(Z'' - w)\big\}\\
&\quad + \sqrt{(P' - p_2P'')}\big\{\tfrac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma}\, p_1\,(Z' - w) - \gamma'\alpha''\beta''(Z'' - w)\big\}\\
&\quad + (p_2 - p_1)\,\mu\alpha'\alpha''\beta'\sigma\sqrt{\cdots}\,(z_0 + \cdots)\, zw = 0.
\end{aligned}$$

In fact the norm of the left-hand side is

$$\begin{aligned}
&(p_1 - p_2)^2(\omega - \varpi)^2\big\{w^2\big[A + 4z(\lambda y + \mu x) - 4\lambda\mu z^2\big]\\
&\quad + 2w\big[B - 2z(\lambda yX + \mu xY)\big]\\
&\quad + C\big\}.
\end{aligned}$$

To partially verify this, observe that, writing the equation under the form $w\sqrt{R} + \sqrt{S} + \sqrt{T} = 0$, on writing therein $w = 0$, we ought to have

$$(R - S)^2 = (p_1 - p_2)^2(\omega - \varpi)^2(\alpha\alpha'\alpha''yz + \beta\beta'\beta''zx + \gamma\gamma'\gamma''xy)^2.$$

But writing $w = 0$, we have

$$\begin{aligned} R - S &= (P' - p_2 P'')\,(\gamma'\alpha'\beta'' p_1 Z' - \gamma'\alpha''\beta'' Z'') \\ &\quad - (P' - p_1 P'')\,(\gamma'\alpha'\beta'' p_2 Z' - \gamma'\alpha''\beta'' Z'') \\ &= (p_2 - p_1)\,(\gamma'\alpha''\beta'' P'' Z' - \gamma'\alpha''\beta'' P' Z''), \\ &= (p_2 - p_1)\big[\gamma''(\beta'\gamma' x + \gamma'\alpha'' y + \alpha'\beta' z)\,(\beta\gamma\beta x - \alpha\alpha'' y) \\ &\quad - \gamma'(\beta''\gamma'' x + \gamma''\alpha'' y + \alpha''\beta'' z)\,(\beta\beta' x - \alpha\alpha' y)\big], \end{aligned}$$

which is easily found to be

$$= -(p_2 - p_1)(\omega - \varpi)(\alpha\alpha'\alpha'' yz + \beta\beta'\beta'' zx + \gamma\gamma'\gamma'' xy);$$

and thus $(R - S)^2$ has the value in question.

171. In further verification, observe that, writing $x, y, z = \alpha'\alpha'', \beta'\beta'', \gamma'\gamma''$, and therefore $P' = 0$, $P'' = 0$, we ought to have

$$\tfrac{1}{4}(p_1 - p_2)^2(\alpha'\beta'\gamma')^2\Big(\cdots\Big)z^2w^2 = (p_1 - p_2)^2(\omega - \varpi)^2 w^2\big[A + 4z(\lambda y + \mu x) - 4\lambda\mu z^2\big],$$

observing that, for the values in question, B, X, Y, C all vanish.

This is

$$(p_1 - p_2)^2(\alpha'\beta'\gamma')^2\Big(\tfrac{\lambda}{\alpha} + \tfrac{\mu}{\beta}\Big)(\gamma'\gamma'')^2 z^2 w^2 = (\omega - \varpi)^2\big[(\omega - \varpi)^2 + 4\gamma'\gamma''(\lambda\beta'\beta'' + \mu\alpha'\alpha'' - \lambda\mu\gamma'\gamma'')\big],$$

which is in fact the value of $(p_1 - p_2)^2$ obtained from the equation in p.

The $13(\alpha)$-nodal Surface $3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$.

172. The equation, node-form $(4_3, 1, 1)$, is

$$w^2\big\{A + 4(\lambda yz + \mu zx + \nu xy) - 4(\mu\nu x^2 + \nu\lambda y^2 + \lambda\mu z^2)\big\}$$
$$+2w\big\{B - 2(\lambda yzX + \mu zxY + \nu xyZ)\big\} + C = 0;$$

for the circumscribed cone we have

$$\begin{aligned} &\big\{A + 4(\lambda yz + \mu zx + \nu xy) - 4(\mu\nu x^2 + \nu\lambda y^2 + \lambda\mu z^2)\big\}\, C \\ &\quad - \big\{B - 2(\lambda yzX + \mu zxY + \nu xyZ)\big\}^2 \\ &= 4\big\{\lambda\alpha yz^2 + \mu\beta zx^2 + \nu\gamma xy^2 + \beta\gamma\alpha xy^2 + \gamma\alpha\beta yz^2 + \alpha\beta\gamma z^2 xy\big\} \\ &\quad \cdot \Big\{-\lambda\frac{X^2}{\alpha^2} - \mu\frac{Y^2}{\beta^2} - \nu\frac{Z^2}{\gamma^2} + \frac{1}{\alpha'\alpha''\beta'\beta''\gamma'\gamma''}\,P'P''\Big\}, \end{aligned}$$

where, on the right-hand side, the first factor, equated to zero, represents a trinodal quartic cone, the nodal lines whereof are $(y = 0, z = 0)$, $(z = 0, x = 0)$, $(x = 0, y = 0)$.

173. As regards the second factor, it is to be observed that, writing as above

$$\omega - \varpi = \frac{\beta'\gamma'' - \beta''\gamma'}{\alpha} = \frac{\gamma'\alpha'' - \gamma''\alpha'}{\beta} = \frac{\alpha'\beta'' - \alpha''\beta'}{\gamma},$$

we have identically

$$\alpha' P' - \alpha'' P'' = \frac{\cdots}{\beta'\beta''}\,x - \cdots,$$

$$\beta' P' - \beta'' P'' = \cdots, \quad \gamma' P' - \gamma'' P'' = \cdots,$$

so that the second factor is

$$\tfrac{1}{(\omega-\varpi)^2}\Big[-\lambda\big(\beta'\beta''\gamma'\gamma''\big)^2(\alpha'P'-\alpha''P'')^2-\mu\big(\gamma'\gamma''\alpha'\alpha''\big)^2(\beta'P'-\beta''P'')^2$$
$$-\nu\big(\alpha'\alpha''\beta'\beta''\big)^2(\gamma'P'-\gamma''P'')^2+(\omega-\varpi)^2\,\alpha'\alpha''\beta'\beta''\gamma'\gamma''P'P''\Big];$$

or, what is the same thing,

$$\tfrac{1}{(\omega-\varpi)^2}\,\alpha'\alpha''\beta'\beta''\gamma'\gamma''\Big\{\cdots(\alpha'\beta'\gamma'P')^2+\cdots(\alpha''\beta''\gamma''P'')^2+2[\cdots]\,\alpha'\beta'\gamma'P'\,\alpha''\beta''\gamma''P''\Big\},$$

so that, equating to zero, we have a pair of planes, each passing through the line $P'=0$, $P''=0$, and which it is clear must be the tangent planes from this line to the quadric cone $A'=0$. I will presently return to this, but I consider first the foregoing identical equation in regard to the circumscribed cone.

174. In verification hereof, observe first that, if $x=0$, the equation becomes

$$\begin{aligned}&\big\{(y-z)^2+4\lambda yz-4\lambda(\beta\gamma y^2+\mu z^2)\big\}\,(\alpha\alpha'\alpha'')^2y^2z^2\\&\quad-\big[(\alpha\alpha'\alpha'')(y^2z-yz^2)-2\lambda yz\alpha(\beta'\beta''y-\gamma'\gamma''z)\big]^2\\&=4\lambda y^2z^2\big[-\lambda\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2-(\alpha\alpha'\alpha'')^2(\nu y^2+\mu z^2)-\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z)\big],\end{aligned}$$

viz., omitting terms which destroy each other, this is

$$\big[(y-z)^2+4\lambda yz\big](\alpha\alpha'\alpha'')^2y^2z^2-\big[\alpha\alpha'\alpha''(y^2z-yz^2)-2\lambda yz\,\alpha(\beta'\beta''y-\gamma'\gamma''z)\big]^2$$
$$=4\lambda y^2z^2\big[-\lambda\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2-\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z)\big].$$

Or again, this is

$$4\lambda yz\,(\alpha\alpha'\alpha'')^2y^2z^2+4\lambda yz\,\alpha^2\alpha'\alpha''(y^2z-yz^2)(\beta'\beta''y-\gamma'\gamma''z)$$
$$-4\lambda^2y^2z^2\alpha^2(\beta'\beta''y-\gamma'\gamma''z)^2=-4\lambda yz\,\alpha^2\alpha'\alpha''(\gamma'y+\beta'z)(\gamma''y+\beta''z);$$

viz., this is

$$\alpha'\alpha''yz+(y-z)(\beta'\beta''y-\gamma'\gamma''z)=(\gamma'y+\beta'z)(\gamma''y+\beta''z),$$

which is at once seen to be true.

175. Again, compare the terms which contain x^2yz. On the right-hand side, we have

$$\big(-K\nu-\mu-\nu-\tfrac14\cdots\big)4\beta\gamma yz\ \cdots\ \text{(a term in } x^2yz \text{ of } \alpha'\alpha''\beta\beta'\gamma\gamma''P'P'')^2,$$

viz., the coefficient is

$$4\beta\gamma\big\{-\mu(\gamma''\gamma)^2-\nu(\beta'\beta'')^2+\beta'\beta''\gamma'\gamma''\big\}.$$

On the left-hand side, that is in $A'C-B'^2$, the only terms which give rise to the terms in question are

$$\text{in } A',\ (1-4\mu\nu)x^2;\ \text{in } C,\ 2\beta\beta'\beta''\gamma\gamma'\gamma''\,x^2yz;$$

and in B^2

$$x^2z\big(-\gamma\gamma'\gamma''-2\mu\gamma'\beta'\beta''\big)-x^2y\big(\beta\beta'\beta''-2\nu\beta\gamma'\gamma''\big),$$

whence the coefficient is

$$-2\beta\beta'\beta''\gamma\gamma'\gamma''(1-4\mu\nu)+2\big(\gamma\gamma'\gamma''+2\mu\beta'\beta''\big)\big(\beta\beta'\beta''+2\nu\beta\gamma'\gamma''\big),$$

which is in fact

$$=4\beta\gamma\big\{-\mu(\gamma'\gamma'')^2-\nu(\beta'\beta'')^2+\beta'\beta''\gamma'\gamma''\big\},$$

which is right; and the verification may be completed without difficulty.

176. The singular planes are

$$\begin{array}{ll} w = 0, & (4) \\ P' - p_1 P'' = 0, & (12) \\ P' - p_2 P'' = 0, & (16) \\ z = 0, & (3) \end{array}$$

and the nodes are

$$\begin{array}{lr} (0,0,0,1), & (8) \\ (1,0,0,0), & (5) \\ (0,1,0,0), & (6) \\ (0,0,1,0), & (7) \\ \left(\frac{\alpha\alpha'\alpha''}{\alpha'-\alpha p_1},\ \frac{\beta\beta'\beta''}{\beta'-\beta p_1},\ \frac{\gamma\gamma'\gamma''}{\gamma''-\gamma p_1},\ 0\right), & (16) \\ \left(\frac{\alpha\alpha'\alpha''}{\alpha'-\alpha p_2},\ \frac{\beta\beta'\beta''}{\beta'-\beta p_2},\ \frac{\gamma\gamma'\gamma''}{\gamma''-\gamma p_2},\ 0\right), & (12) \\ (\alpha'\alpha'',\ \beta'\beta'',\ \gamma'\gamma'',\ 0), & (4) \\ \text{three nodes on line } P' - p_1 P'' = 0,\ z = 0 & (13, 14, 15) \\ \text{three nodes on line } P' - p_2 P'' = 0,\ z = 0 & (9, 10, 11) \end{array}$$

where the large numbers are those referring to the 16-nodal surface, and are here adopted. In the foregoing formulae p_1, p_2 are the roots of

177.

$$\begin{aligned} &\left[\lambda(\beta'\gamma'')^2 + \mu(\gamma'\alpha')^2 + \nu(\alpha'\beta')^2\right](p\alpha'\beta'\gamma')^2 \\ &\quad + \left[\lambda(\beta''\gamma'')^2 + \mu(\gamma''\alpha'')^2 + \nu(\alpha''\beta'')^2\right](\alpha''\beta''\gamma'')^2 \\ &\quad - 2\left[\lambda\beta'\beta''\gamma'\gamma'' + \mu\gamma'\alpha'\gamma''\alpha'' + \nu\alpha'\beta'\alpha''\beta'' + \tfrac{1}{4}(\omega - \varpi)^2\right] p\alpha'\beta'\gamma'\alpha''\beta''\gamma'' = 0. \end{aligned}$$

We have the square diagram in the following page:

[The original page 296 here gives a 13×13 square diagram for the $13(\alpha)$-nodal surface, with notations including the cubic cone C (or G) replacing the planes $5, 6, 7$ at node 4, the trinodal quartic cone $(8;\ 1, 2, 3)$ at node 8, the nodal cubic $(2, 3;\ 5)$ at node 5, &c. The vacant places are understood to be CC. See PDF page 296 for the full table.]

where for greater clearness I have omitted the symbol CC, which is to be understood as occupying each of the vacant squares.

The arrangement is the same as before; the right-hand margin shows the sextic cone; viz., for the node 4 this is made up of the singular planes 4, 12, 16 and of a cubic cone represented by $(5, 6, 7)$ (as replacing the planes 5, 6, 7 in the case of the 16-nodal surface). Similarly for the node 5, the sextic cone is made up of the singular plane 4, the nodal cubic cone $(2, 3;\ 5)$, and the quadric cone $(9, 13)$ (the numbers in these last symbols indicating the planes in the case of the 16-nodal surface, which are here replaced by cones). So for the node 8, the sextic cone is made up of the singular planes 12, 16 and of the trinodal quartic cone $(8;\ 1, 2, 3)$. As regards a nodal cubic cone, for example $(2, 3;\ 5)$, the semicolon is used to indicate that the nodal line replaces the intersection of the planes 2, 3; the other intersections 2, 5 and 3, 5 having disappeared. And so for a trinodal quartic cone $(8;\ 1, 2, 3)$, the semicolon is used to indicate that the nodal lines replace the intersection $(1, 2)$, the intersection $(1, 3)$, and the intersection $(2, 3)$ respectively; the other intersections 1, 8; 2, 8; and 3, 8 having disappeared. Finally, in the body of the table, C

is used to denote the cubic or the quartic cone (as the case may be); $\times$ to denote a nodal line of either of these cones; and C' the quadric cone; as already mentioned, the vacant places are considered to be CC.

The reading of the table is then as follows; viz., for the node 4, the remaining twelve nodes lie on the nodal lines of the sextic cone 4, 12, 16, $(5, 6, 7)$, as shown; viz., 5, 6, 7 are each of them on the intersection of the cubic cone with the plane 4; 8 is on the intersection of the planes 12 and 16; and so on.

I reserve for another Memoir the discussion of the $13(\beta)$-nodal surface, and the surfaces with less than 13 nodes.

[455] On Plücker's Models of Certain Quartic Surfaces.

[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 281–285. *Read June* 8, 1871.*]*

The Society possesses a series of 14 wooden models of surfaces, constructed under the direction of the late Prof. Plücker, in illustration of the theory developed in his posthumous work, "Neue Geometrie des Raumes gegründet auf die Betrachtung der geraden Linie als Raumelemente," Leipzig, 1869. These all of them represent, I believe, Equatorial Surfaces; viz., models 1 to 8 represent cases of the 78 forms of equatorial surfaces "deren Breiten-Curven eine feste Axenrichtung besitzen," vol. ii. pp. 352–363, the remaining models, Nos. 9–14, I have not completely identified. I propose to go into the theory only so far as is required for the explanation of the models.

In a "Complex," or triply infinite system of lines, there is in any plane whatever a singly infinite system of lines enveloping a curve; and if we attend only to the curves the planes of which pass through a given fixed line, the locus of these curves is a "complex surface." Similarly, there is through any point whatever a single infinite series of lines generating a cone; and if we attend only to the cones which have their vertices in the given fixed line, then the envelope of these cones is the same complex surface. In the case considered of a complex of the second degree, the curves and cones are each of them of the second order; the fixed line is a double line on the surface, so that (attending to the first mode of generation) the complete section by any plane through the fixed line is made up of this line twice, and of a conic; the surface is thus of the order 4: it is also of the class 4; the surface has, in fact, the nodal line, and also 8 nodes (conical points), and we have thus a reduction $= 32$ in the class of the surface.

In the particular case where the nodal line is at infinity, the complex surface becomes an equatorial surface; viz., (attending to the first mode of generation) we have here a series of parallel planes each containing a conic, and the locus of these conics is the equatorial surface.

It is convenient to remark that, taking a, b, h to be homogeneous functions of (x, w) of the order 2; f, g of the order 1; and c of the order 0 (a constant); then the equation of a complex surface (see vol. i. p. 162) is

$$\begin{vmatrix} \cdot, & y, & z, & 1 \\ y, & a, & h, & g \\ z, & h, & b, & f \\ 1, & g, & f, & c \end{vmatrix} = 0;$$

and that, writing $w = 1$, or considering a, h, b; f, g; c as functions of x of the orders 2, 1, 0 respectively, we have an equatorial surface.

A particular form of equatorial surface is thus $bcy^2 + caz^2 + ab = 0$, or taking $c = 1$, this is $by^2 + az^2 + ab = 0$, where a, b are quadric functions of x.

The surface is still, in general, of the fourth order: it may however degenerate into a cubic surface, or even into a quadric surface; the last case is however excluded from the enumeration. The section by any plane parallel to that of yz is a conic; the section by the plane $y = 0$ is made up of the pair of lines $a = 0$, and of the conic $z^2 + b = 0$; that by the plane $z = 0$ is made up of the pair of lines $b = 0$, and of the conic $y^2 + a = 0$; the last-mentioned planes may be called the principal planes, and the conics contained in them principal conics. The surface is thus the locus of a variable conic, the plane of which is parallel to that of yz, and which has for its vertices the intersections of its plane with the two principal conics respectively. And we have thus the particular equatorial surfaces considered by Plücker, vol. ii. pp. 346–363 (as already mentioned), under the form

$$\frac{y^2}{Ex^2 + 2Ux + G} + \frac{z^2}{Fx^2 + 2Rx + B} + 1 = 0,$$

and of which he enumerates 78 kinds; viz., these are

1 to 17 Principal conics each proper.

18 „ 29 One of them a line-pair.

30 „ 32 Each a line-pair.

33 „ 39 Principal conics, each proper, but having a common point.

40 „ 43 One of them a line-pair, its centre on the other principal conic.

44 „ 61 One principal conic a parabola.

62 „ 73 One principal conic a pair of parallel lines.

74 „ 76 Principal conics each a parabola.

77 and 78 Principal conics, one of them a parabola, the other a pair of parallel lines.

The models 1 to 8 correspond hereto, as follows:

Mod.	1 is	13,	i.e.	2 is Mod.	9,
„	2	9,	„	3	40,
„	3	40,	„	4	34,
„	4	34,	„	5	2,
„	5	2,	„	6	3,
„	6	3,	„	7	4,
„	7	4,	„	8	32,
„	8	32,	„	13	1.

Mod. 5 is 2: the form of the equation is here

$$\frac{y^2}{l^2\left[(x-\alpha)^2 + \beta^2\right]} + \frac{z^2}{l'^2\left[(x-\alpha')^2 + \beta'^2\right]} = 1;$$

viz., the principal conics are one of them a hyperbola, the other imaginary; hence the generating conic has always two, and only two, real vertices, viz., it is always a hyperbola; there are no real lines.

Mod. 6 is 3: the form of the equation is

$$\frac{y^2}{l^2\left[(x-\alpha)^2 + \beta^2\right]} + \frac{z^2}{l'^2\left[(x-\alpha')^2 + \beta'^2\right]} = 1;$$

viz., the principal conics are each of them a hyperbola; the generating conic has four real vertices, viz., it is always an ellipse: there are no real lines.

Mod. 7 is 4: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2[(x-\alpha')^2+\beta'^2]} + 1 = 0.$$

The principal conics are one of them an ellipse, the other imaginary; for values of x between γ and δ, the variable conic has two real vertices or it is a hyperbola; for any other values it is imaginary, so that the surface lies wholly between the planes $x=\gamma$, $x=\delta$: the surface contains the real lines $y=0$, $x=\gamma$ and $y=0$, $x=\delta$.

Mod. 2 is 9: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0,$$

where, say the values γ, δ lie between the values γ', δ': the principal conics are each of them an ellipse, the vertices (on the axis or line $y=0$, $z=0$) of the one ellipse lying between those of the other ellipse. The variable conic for values of x between γ, δ has four real vertices, or it is an ellipse; for values beyond these limits, but within the limits γ', δ', say from γ to γ', and from δ to δ', there are two real vertices, or the conic is a hyperbola; and for values beyond the limits γ', δ', the variable conic is imaginary.

There are four real lines $(y=0, x=\gamma)$, $(y=0, x=\delta)$, $(z=0, x=\gamma')$, $(z=0, x=\delta')$. The surface consists of a central pillow-like portion, joined on by two conical points to an upper portion, and by two conical points to an under portion, the whole being included between the planes $x=\gamma'$, $x=\delta'$.

Mod. 1 is 13: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0;$$

the values γ', δ' lying between γ, δ; the principal conics are one of them a hyperbola, the other an ellipse, the vertices (on the axis or line $y=0$, $z=0$) of the hyperbola lying between those of the ellipse.

The variable conic, for values of x between γ', δ', has two real vertices, or it is a hyperbola; for the values, say, from γ' to γ, and δ' to δ, there are four real vertices, or the conic is an ellipse; for values beyond the limits γ, δ, there are two real vertices, and the conic is a hyperbola. There are the four real lines $(y=0, x=\gamma)$, $(y=0, x=\delta)$, and $(z=0, x=\gamma')$, $(z=0, x=\delta')$. The surface consists of 8 portions joined to each other by 8 conical points, but the form can scarcely be explained by a description.

Mod. 8 is 32: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)^2} + \frac{z^2}{l'^2(x-\gamma')^2} = 1;$$

viz., the principal conics are each of them a line-pair, the variable conic is always an ellipse.

There are the two real nodal lines $(y=0, x=\gamma)$ and $(z=0, x=\gamma')$, each of these being in the neighbourhood of the axis crunodal, and beyond certain limits acnodal; the surface is a scroll, being, in fact, the well-known surface which is the boundary of a small circular pencil of rays obliquely reflected, and consequently passing through two focal lines.

Mod. 4 is 34: the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta)} + 1 = 0,$$

where $x = \delta$ is not intermediate between the values $x = \gamma$ and $x = \gamma'$; say the order is δ, γ, γ'. The surface is thus a cubic surface; the principal conics are ellipses having on the axis a common vertex at the point $x = \delta$, and the remaining two vertices on the same side of the last-mentioned one. The variable conic for values between δ and γ has four real vertices, or it is an ellipse; for values between γ and γ' two real vertices, or it is a hyperbola; and for values beyond the limits δ, γ' it is imaginary. There are on the surface the two real lines $(y = 0, x = \gamma)$ and $(z = 0, x = \gamma')$. The surface consists of a finite portion joined on by two conical points to the remaining portion.

Mod. 3 is 40: the form of equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\delta)^2} + 1 = 0.$$

The surface is thus a cubic surface; the principal conics are, one of them an ellipse, the other a pair of imaginary lines intersecting on the ellipse; for values of x between γ and δ, the variable conic has thus two real vertices, and it is a hyperbola; for values beyond these limits it is imaginary, and the whole surface is thus included between the planes $x = \gamma$ and $x = \delta$. There are the two real lines $(y = 0, x = \gamma)$ and $(z = 0, x = \delta)$.

Taking $l^2 = l'^2 = 1$, the surface is

$$\frac{y^2}{(x-\gamma)(x-\delta)^2(x-\delta)^2} + \frac{z^2}{\cdots} + 1 = 0,$$

which is a particular case of the parabolic cyclide.

As already mentioned, I have not completely identified the remaining models 9 to 14, but I will say a few words about them.

The equatorial surfaces, not included in the preceding 78 cases, Plücker distinguishes (vol. ii. p. 363) as “gedrehte” or “tordirte,” say as twisted equatorial surfaces; the equation of such a surface is

$$by^2 + 2hyz + az^2 + ab - h^2 = 0,$$

where

$$\begin{aligned} b &= Fx^2 - 2Rx + B, \\ a &= Ex^2 + 2Ux + G, \\ h &= Kx^2 - Ox - G \text{ (or in particular } = -Ox - G). \end{aligned}$$

Mod. 13 is such a surface, being a twisted form of model 2.

Mod. 9 and Mod. 14 belong, I think, to the case $a = 0$; viz., the form of the equation is $by^2 + 2hyz - h^2 = 0$. The variable conic is a hyperbola, the direction of one of the asymptotes being constant (vol. ii. p. 368).

There are moreover (p. 372) equatorial surfaces in which the variable conic is always a parabola, and where there are on the surface four real or imaginary lines.

Mods. 10, 11, and 12 seem to represent such surfaces.

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[456] NOTE ON THE DISCRIMINANT OF A BINARY QUANTIC.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), p. 23.]

It is well known that the discriminant of a binary quantic $(a, b, c, d, \ldots | t, 1)^n$ is of the form

$$Ma + Nb^2,$$

but it is further to be remarked that if $b = 0$, then the form is

$$a(Ma + Nc^3),$$

if $b = 0$, $c = 0$, the form is

$$a^2(Ma + Nd^4),$$

and so on, until only the lowest two coefficients are not put $= 0$. Or, what is the same thing, if in the discriminant of the original function we put $a = 0$, then the discriminant divides by b^2; if $b = 0$, the discriminant divides by a, and, omitting this factor, if we then write $a = 0$, it divides by c^3; if $b = 0$, $c = 0$, the discriminant divides by a^2, and omitting this factor, if we then write $a = 0$, it divides by d^4; and so on, until as before.

Thus if $b = 0$, the discriminant of $(a, 0, c, d, e \,|\, t, 1)^4$, divides by a, and omitting this factor it is

$$\begin{aligned} & a^2e^2 \\ & -\ 18\,ace^2 \\ & +\ 54\,acd^2e \\ & -\ 27\,ad^4 \\ & +\ 81\,c^4e \\ & -\ 54\,c^3d^2, \end{aligned}$$

which for $a = 0$ has the factor c^3; if $b = 0$, $c = 0$, the discriminant of $(a, 0, 0, d, e \,|\, t, 1)^4$ has the factor a^2, and omitting this factor it is

$$ae^4 - 27\,d^4,$$

which for $a = 0$ has the factor d^4; the series of theorems here terminates, since the lowest two coefficients d, e are not to be put $= 0$.

(End of chunk: PDF pages 276–325, book pages 254–303.
Paper 450 (tail, two pages),
paper 451 in full,
paper 452 in full,
paper 453 in full,
paper 454 in full,
paper 455 in full,
paper 456 opening (one page).)